COMPETITIVE INTELLIGENCE & FUNDING ANALYSIS REPORT

HCSP Competitor Intelligence — Plain-English Edition

Four verified SCWO competitor profiles — 374Water, General Atomics EMS, Aquarden Technologies, Revive Environmental — four customer-and-geography deep dives, and 35 screened non-dilutive funding channels.

The full competitive picture in plain words: who can sell to small and mid-size facilities today, who cannot and why, and where the money is.

Prepared September 2026

Table of Contents

Executive summary: the four-company verdict in plain words 3

1. 374Water Inc. (product brand: AirSCWO) 4

1.1 Financial health 4

1.2 Customers: real ones vs. announced hopes 6

1.3 How small can 374Water go — and who it sells to 7

2. General Atomics EMS (product brand: PERSES) 7

2.1 Who stands behind it 7

2.2 Track record: 6 million gallons destroyed — and a checkered history 8

2.3 How small can GA go — and who it sells to 9

3. Aquarden Technologies (product brand: SuperOx) 9

3.1 Financial health: family-owned, loss-making, worth less than nothing on paper 10

3.2 Customers: two real industrial systems, one controlling partner 11

3.3 How small can Aquarden go — and who it sells to 12

4. Revive Environmental (product: Battelle PFAS Annihilator) 12

4.1 Who stands behind it 13

4.2 Customers: four state collection programs, one state lost, two fixed plants 13

4.3 How small can Revive go — and who it sells to 14

5. Customers and Where They Operate 14

5.1 374Water’s customers 15

5.2 General Atomics’ customers and hosts 16

5.3 Aquarden’s customers and test hosts 17

5.4 Revive’s customers 18

5.5 What the map shows 20

6. The Scale Gap — Why Small Facilities Are Left Out, and What HCSP Should Do About It 20

6.1 The four competitors side by side 20

6.2 The evidence for the gap 21

6.3 What HCSP must do to hold the opening 21

7. Money That Doesn’t Cost Equity: Grants, Prizes, and Partner Channels 22

7.1 Channels HCSP can apply to directly 22

7.2 Channels reachable through a partner 25

7.3 Ruled out — so no one wastes proposal effort 26

7.4 How this stacks onto the existing U.S. register 26

The Words We Use (One-Page Glossary) 26

References 28

A quick note on two terms used throughout: ***supercritical water oxidation (SCWO)*** *is a way of destroying waste with very hot, high-pressure water — hot enough and squeezed hard enough that the waste chemically burns up in the water itself.* ***PFAS "forever chemicals"*** *are a family of industrial chemicals that do not break down in nature; destroying them is the main reason customers buy this kind of equipment.*

Executive summary: the four-company verdict in plain words

Not one credible SCWO competitor currently sells to small and mid-size facilities — and each of the four is blocked from doing so by the way it is built, funded, or organized. That is the single most important finding in this report, and it means the part of the market HCSP wants — small factories, farms, food and drink plants, and remote sites that want waste destroyed on their own premises — is effectively unserved today.

Each competitor fails this market in a different way:

- **374Water** (the only stock-market-listed company focused purely on SCWO) is running out of money. In its 2025 fiscal year it took in $215,037 of revenue while losing $21.0 million.[^W-2^] It had $1.78 million in cash on June 30, 2026,[^W-6^] and its own auditors warn in two consecutive filings that the company may not be able to stay in business without new money (this warning is called a "going-concern" note).[^W-3^][^W-6^]
- **General Atomics**, a roughly $3.2 billion-a-year private defense company, has the oldest and best-proven SCWO technology (branded PERSES). But the waste line is a tiny side business inside a weapons and space company. Everything it has done in the U.S. in 2025–2026 has been a government-paid demonstration, not a commercial sale, and we could find no publicly verifiable permanent installation at a paying American customer.[^G-13^][^G-27^]
- **Aquarden Technologies** (Denmark) has the one thing nobody else can prove — industrial SCWO plants actually running for customers in France — but its finances are the weakest of all: it owes more than it owns (negative net worth of −DKK 30.9 million, about −$4.8 million, at the end of 2025), it has lost money every year from 2021 through 2025, and it employs only 11 people.[^A-1^] Almost everything it does flows through one customer, the French aerospace group ArianeGroup.
- **Revive Environmental** (a joint venture of the nonprofit research giant Battelle and the hedge fund Viking Global) is the best-funded and best-permitted player in North America — but it chose a "ship your waste to us" model. Customers send their waste to Revive's two fixed plants in Ohio and Michigan. Small waste producers only get served when a state government collects waste from many of them at once and trucks it in.[^R-16^][^R-39^]

The smallest deals in this industry tell the story. The cheapest equipment purchase we could verify anywhere in the field is 374Water's $4.88 million project for the city of Olathe, Kansas.[^W-6^] General Atomics' smallest product is a shipping-container unit processing about 3 gallons per minute, with no published price and no service plan.[^G-9^] Aquarden sells one reactor size — about 5 tonnes per day, in the $2–10 million purchase-cost range — and can only grow by bolting more identical reactors side by side.[^A-6^][^A-21^] Revive does not sell machines at all; it charges by the gallon, and the freight costs of shipping waste to the Midwest price out anyone too small to fill a truck.[^R-28^] **Nobody offers an on-site system under $1 million that a small facility buys and runs itself. That gap is HCSP's opening.**

The opening is real but not guaranteed to last. Four things could close it, and HCSP should track them: (1) a Pentagon "Success Memo" — a stamp of approval that turns a government demonstration into big orders — for General Atomics (374Water already received one in August 2026); (2) ArianeGroup absorbing Aquarden's sales channel under its ELIXIR service brand; (3) Revive deciding to put its container units on customer sites instead of requiring shipping; and (4) a rival technology (Aquagga's HALT process) undercutting everyone on price, as it already did when Colorado chose it at $40 per gallon.[^W-28^][^A-27^][^R-24^][^R-25^]

On funding, this report also covers 35 grant, prize, and low-cost-capital channels beyond the existing U.S. register. Eight are open to a U.S. for-profit company like HCSP today — led by the Asian Development Bank's Technology Innovation Challenge (pilot grants up to $450,000), the AB InBev 100+ Accelerator (pilots up to $100,000 with five big beverage companies watching), and WaterEquity's Water & Climate Resilience Fund ($2–15 million investments).[^N-15^][^N-34^][^N-26^] Section 7 explains these in plain terms.

---

1. 374Water Inc. (product brand: AirSCWO)

Bottom line: 374Water is the closest thing HCSP has to a head-to-head SCWO competitor — and it is one or two fundraising events away from serious trouble. Its finances are weak, its biggest contract loses money, and it can only build roughly one system at a time. Even if it wanted to serve lots of small customers, it could not afford to. HCSP should treat 374Water as proof that the technology works and as a "will this vendor still exist in five years?" story to raise in every competitive sales conversation — not as a competitor to match on price.

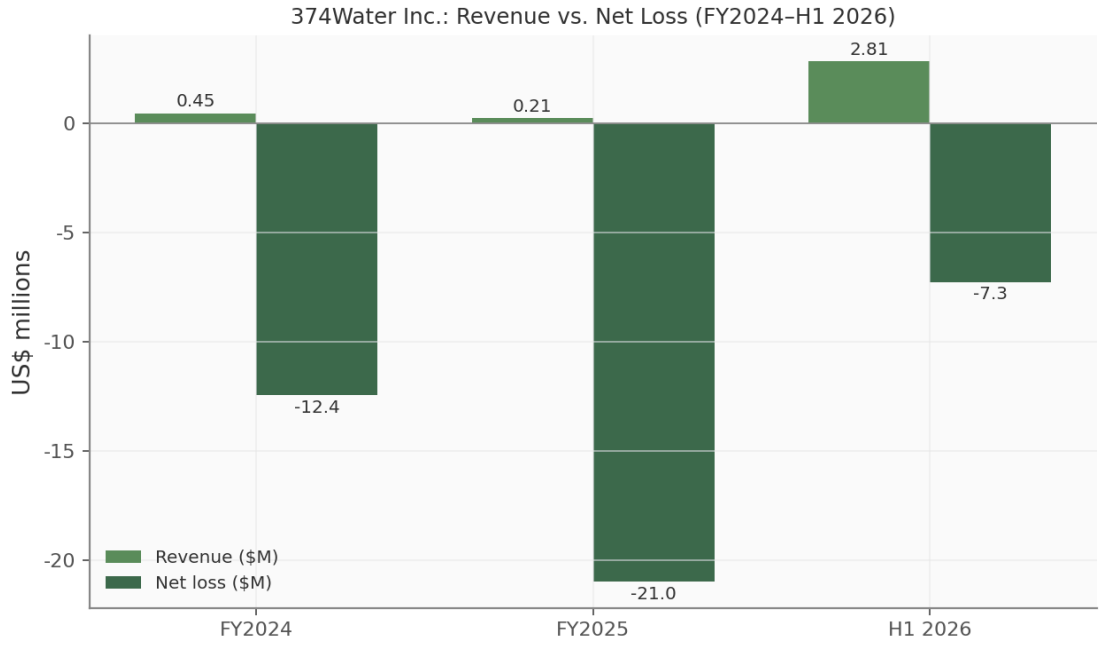
1.1 Financial health

374Water is a small publicly traded company (stock ticker SCWO) based in the Raleigh–Durham area of North Carolina. It was spun out of Duke University in 2018 and went public in 2021 through a merger with an already-listed shell company.[^W-13^][^W-14^][^W-16^] Its stock market value in September–October 2026 was roughly $42–54 million.[^W-16^][^W-17^] The financial picture is one of shrinking revenue and growing losses, briefly interrupted in 2026 by a single one-time event:

Measure	FY2024	FY2025	First half of 2026
Revenue	$445,445 [^W-1^]	$215,037 [^W-2^]	$2,813,195 [^W-6^]
Net loss (money lost after all costs)	$(12,434,114) [^W-2^]	$(20,975,052) [^W-2^]	$(7,267,858) [^W-6^]
Cash burned running the business	$(10,589,735) [^W-2^]	$(14,326,205) [^W-2^]	$(2,297,376) [^W-6^]
Cash in the bank (end of period)	$10,651,644 [^W-2^]	$3,198,682 [^W-2^]	$1,776,259 [^W-6^]
Total losses piled up since founding	$28,387,618 [^W-1^]	$49,936,598 [^W-3^]	$57,204,456 [^W-6^]

All figures are from the company's own SEC filings (annual 10-K reports for FY2024 and FY2025; the quarterly 10-Q for Q2 2026). Note: FY2025 revenue would have looked better but for a roughly $1.9 million negative

adjustment after the company changed how it accounted for its delayed Orange County contract.[^W-2^]



374Water revenue vs. net loss, FY2024 through first half of 2026. Source: 374Water SEC filings (Forms 10-K and 10-Q).[^W-2^][^W-6^]

Read the table this way: the company went into 2024 with about $10.7 million in cash and has burned through nearly all of it in two and a half years. The first-half 2026 revenue "recovery" is misleading — roughly $2.0 million of the $2.8 million came from one event, passing a pre-shipment factory test (called a Factory Acceptance Test) on the long-delayed Orange County machine; first-quarter revenue alone was only $551,155.[^W-2^][^W-5^][^W-8^]

Three warning signs matter more than any single number. **First, the company's ability to sell new shares is nearly shut.** It announced a headline $50 million "at-the-market" share-sale facility in August 2026, but SEC rules for very small companies (the "baby-shelf" rule) cap actual sales at one-third of its roughly $27 million of freely tradable stock — leaving only about $1.14 million of stock it could actually sell under the August 2026 prospectus.[^W-3^][^W-6^][^W-7^] It has instead resorted to borrowing $860,000 through convertible notes (loans that turn into shares) at a 10% interest rate with free share-purchase warrants attached, plus a roughly $630,000 loan secured against company assets.[^W-3^][^W-6^] **Second, the biggest shareholder is selling out.** Co-founder Yaacov Nagar, who owns 18.06% of the company, made 26 separate share sales totaling 1,322,292 shares (about $1.29 million) in six months — selling the company's own 10-K blames for depressing the share price — and the company had to do a 1-for-10 reverse share split in December 2025 just to keep its Nasdaq listing.[^W-3^][^W-9^][^W-12^][^W-18^] **Third, the runway is measured in months.** With $1.78 million of cash against a $2.3 million half-year burn rate, the company spends money faster than it has it — which is exactly why both the FY2025 10-K and the Q2 2026 10-Q carry the auditor's warning that the company may run out of money.[^W-3^][^W-6^]

What this means for HCSP: in every bid where 374Water is also at the table, HCSP should politely make vendor survival a buying criterion — ask for performance bonds (insurance that a vendor finishes the job) that 374Water

1.2 Customers: real ones vs. announced hopes

374Water does have real customers — but they are few, and every dollar of booked revenue comes from a government or government-adjacent buyer:

Customer	What they bought	Value	Status (September 2026)
City of Orlando, FL (Iron Bridge water plant)	2024 demo ($812,000), then a 5-year service deal signed March 2026 where 374Water owns and runs the machine on city land	Demo $812,000; about $482,000/quarter of service revenue; company projects $3–5M/yr at initial scale	**Operating** — the flagship site and only fixed service hub [^W-6^][^W-8^][^W-21^][^W-23^]
Orange County Sanitation District, CA	One AirSCWO 6 machine, sold 2022	About $4.6M to 374Water; the district budgeted $8.7M total in 2021	**About 4 years late.** Passed its factory test June 2, 2026; delivery planned October 2026. **A money-losing contract** — the company set aside a $1.6M loss provision and has spent $1.9M more than it has billed [^W-2^][^W-5^][^W-6^][^W-20^][^W-22^]
City of Olathe, KS (via contractor Garney)	One AirSCWO 6 plus pre-treatment equipment	**$4,880,000 fixed price** ($3.0M for the machine); $2.2M milestone payment received in Q1 2026	**Signed but not yet built** — the largest equipment order in company history [^W-6^][^W-8^]
City of St. Cloud, MN	A crewed mobile pilot (state grant-funded)	$600,000	**Completed September 15, 2026** [^W-24^]
State of North Carolina / UNC	Destruction of old firefighting foam	About $1M initially; up to 29,000 gallons possible	Processing has begun [^W-26^]

Everything beyond this table is announcement, not revenue: a paid Pentagon demonstration at a Detroit hazardous-waste facility (about 900 gallons destroyed over 80 hours, with destruction better than 99.9% verified by engineering firm Arcadis) that earned the company a Pentagon "Success Memo" in August 2026; a demonstration at Peterson Space Force Base; a 2022 Navy award; a hosting agreement with waste company Crystal Clean where no plant has been built; a cooperation memo with Arcadis; and a company claim of "identified demand totaling nearly $1.8BN" that lumps together signed work, demonstrations, and lab studies rather than firm orders.[^W-2^][^W-4^][^W-5^][^W-8^][^W-27^][^W-28^][^W-29^]

The gap between what 374Water announces and what it banks is the point. Five years into selling, it has one running site (Orlando), one equipment sale delivered four years late that loses money (Orange County), one signed-but-unbuilt order (Olathe), and grant-funded pilots. The Orange County story — signed 2022, delivery October 2026, multiple missed dates — is the single most useful competitive fact HCSP owns: it shows buyers what happens when a thinly funded vendor tries a city-sized SCWO project, and it explains why smart buyers now structure these deals as small trials with options to expand, not big commitments.[^W-12^][^W-21^][^W-22^]

1.3 How small can 374Water go — and who it sells to

Only two machine sizes actually exist: the AS1 (about 1 wet tonne per day, mobile; the first unit was finished in 2025–26) and the AS6 (6–10 tonnes per day, the only size with commercial operating hours). The bigger AS30 is still in development with first manufacturing promised "late 2026"; the AS100/AS200 exists only in presentations, described by the company itself as something "to be developed in the future based on customer orders." The company's own 10-K quietly narrows the marketing claim of "1 to over 200 tons per day" to an operating line "designed to scale between six (6) and ten (10) tons per day."[^W-3^][^W-4^][^W-30^]

Three hard facts set the bottom of 374Water's market. **First, the cheapest equipment deal in the whole industry is its ~$4.9 million Olathe project.**[^W-6^] **Second, the nominally small AS1 is not for sale** — 374Water keeps it in its own fleet and sends a crew with it for municipal pilots priced at $0.5–1.5 million a year, aimed at towns of roughly 2,500–5,000+ people.[^W-4^][^W-24^] **Third, the math forces it to go big:** management's own investor deck prices a service site at $1.5–10 million to build and only claims payback at large "hub" scale, and its September 2026 strategy announcement explicitly targets big regional plants that collect sludge from many cities — a wholesale model, not a retail one.[^W-4^][^W-32^] The founder himself has said the smallest container unit was "sized for a community of about 6,000 people" and only becomes a net energy producer at about 30,000-person scale.[^W-31^]

So its customers are what you would expect: big city utilities (Orange County processes about 180 million gallons a day; Orlando 40; Olathe 11), federal defense programs, and hazardous-waste companies acting as hosts.[^W-5^][^W-12^][^W-24^][^W-28^] **The customers it does not serve are exactly HCSP's first market: small industrial producers, farms, food and beverage plants, remote sites, and any private buyer without a grant.** No named private industrial customer appears anywhere in its filings or press coverage, and the company's original Gates Foundation mission of small-scale sanitation has been quietly dropped.[^W-2^][^W-4^][^W-14^] With 48 employees, one factory, and two systems in existence at the end of 2025, this is not a marketing choice — it is a question of money and bandwidth.[^W-3^][^W-17^]

---

2. General Atomics EMS (product brand: PERSES)

Bottom line: General Atomics (GA) is the industry's technology grandfather — and its commercial ghost. Thirty-six years of development, roughly $250 million invested, an EPA-verified destruction rate above 99.99% for PFAS, and more than 6 million gallons destroyed make PERSES the "safe" choice when a big government program shops for SCWO. But the product line is a tiny side business inside a ~$3.2 billion private defense contractor; its 2025–2026 U.S. activity is 100% government-funded demonstrations; and — most striking — the two biggest government programs that ever bought its SCWO systems both ended up not using them, citing capacity, reliability, and cost. GA is a competitor for giant national programs only. For HCSP it is less a sales rival than a cautionary tale to quote.

2.1 Who stands behind it

GA has been owned by the Blue family since 1986. It earns about $3.2 billion a year (2024, per Forbes/FactSet), holds $41.5 billion in cumulative federal contract obligations — including a single $14.1 billion drone agreement

awarded September 2025 — and employs around 15,000 people.[^G-1^][^G-3^][^G-4^] The SCWO line sits inside the GA-EMS division, whose real identity is aircraft-carrier launch systems, space, hypersonics, and directed energy; hazardous waste is a stated but marginal tail of the business.[^G-8^][^G-14^] Funding has been a mix of internal research money and Pentagon program money over ~36 years — GA's own 2016 presentation cites "22 years and $250M of SCWO development… over 15 working SCWO systems… over 35,000 hours of steady state operations."[^G-9^][^G-10^] PERSES revenue is not broken out and is, by any reasonable inference from documented sales, at most a few million dollars a year — a rounding error for the parent.

The strategic read for HCSP: PERSES will never die for lack of money — the Blue family's self-funding culture is famous — but one named business-development director carries the entire public effort, the product was only rebranded in 2025 (trademark registered January 2026), and it is marketed at defense conferences rather than waste-industry events.[^G-10^][^G-16^][^G-32^] The risk to a GA customer is not that the vendor goes bankrupt — the opposite of the 374Water and Aquarden situations — but that the project gets deprioritized, delivered slowly, or dropped when a defense program ends. HCSP cannot outspend GA; it can out-iterate, out-price, and out-service it in segments GA will never staff.

2.2 Track record: 6 million gallons destroyed — and a checkered history

GA's verified claims are genuine: more than 6 million gallons of waste destroyed at better than 99.99% efficiency across 200+ waste types, and the first-ever EPA test-and-verification of a commercial SCWO system destroying PFAS in firefighting foam (EPA report EPA/600/R-22/257, September 2022).[^G-15^][^G-16^][^G-17^] The deployment record behind those claims is mixed:

Site / program	What happened	Evidence
Tooele Army Depot, UT — SCWO added to an Army disposal line	**Abandoned**	EPA (2019): the unit "eventually was not used due to lack of capacity and reliability issues, and offsite treatment methods were found to be more cost effective" [^G-19^]
Blue Grass, KY — four-reactor SCWO for the chemical-weapons destruction program	**Built, never operated; rejected August 2020**	The liquid waste was trucked to Veolia in Texas instead; the National Research Council had flagged salt-plugging, corrosion, and long-run reliability concerns as far back as 1999–2002 [^G-11^][^G-20^][^G-21^]
Camp Minden, LA — 3 gallons/minute unit for a private explosives recycler	**Collapsed**	The customer (Explo Systems) went bankrupt; the unit sits stranded [^G-9^]
Kyrgyzstan & Tajikistan — transportable units, UN-funded	Operating / recent	Destroying DDT pesticide in 5,000 tonnes of waste under UN/Basel Convention guidelines [^G-23^]
Tianjin, China — three systems sold (2018)	Follow-through unknown	No public evidence the systems were ever commissioned or reordered [^G-24^]

First commercial sale, 2012–2014 — customer never named	“First two systems are still in use” (GA, Oct 2025)	Buyer’s identity never disclosed [^G-10^]

Current U.S. activity is demonstrations, not operations. The Pentagon-funded demo at a Charlotte, NC waste facility ran six PFAS waste streams in August–September 2025; a U.S. Army Corps of Engineers-funded demo via contractor Bay West ran three streams at Lake Elmo, MN in May–June 2025 before the unit was moved on. As of September 2026, **no public results report or Pentagon “Success Memo” for GA could be found** — a telling contrast, because 374Water received its Success Memo in August 2026.[^G-13^][^G-25^][^G-27^][^G-28^][^W-28^] And one clarification for the record: the full-scale SCWO unit running in France for the defense industry belongs to **Aquarden**, not GA.[^G-33^]

The “so what” cuts both ways. GA’s EPA verification and 36-year pedigree make it the supplier to beat in big Pentagon and federal channels. But Tooele and Blue Grass are government-published evidence that GA’s SCWO was twice rejected by its own anchor customer type — the U.S. Army — on capacity, reliability, and cost grounds. That “graveyard pattern” is why every SCWO buyer is skeptical today, and why HCSP’s reliability documentation (Section 6.3) is a weapon, not paperwork.

2.3 How small can GA go — and who it sells to

GA’s baseline product is a 3 gallon-per-minute container unit (about 4,300 gallons a day) — 25 feet long, 8 feet wide, 9.5 feet tall, roughly 20,800 pounds — running at 650 °C and 4,000 psi and needing a train of supporting equipment (air compression, a caustic-soda quench, and pre- and post-processing).[^G-9^][^G-10^] Bigger fixed units existed historically, but there is **no product smaller than 3 gpm, no published price, no per-gallon service, and no lease or financed offer**.[^G-9^][^G-22^] Real-world deployments need either a government sponsor or a host site with a federal hazardous-waste permit; the Basel Convention’s own guidelines note that SCWO units “are predominantly used in fixed configurations.”[^G-13^][^G-23^] GA’s own marketing positions PERSES against “lab and prototype systems” — i.e., it competes on industrial scale and maturity, which is a declaration that the small end is not its market.[^G-16^]

Its customers confirm the shape: Pentagon environmental programs, the Army Corps of Engineers working through environmental contractors, big hazardous-waste facilities acting as demonstration hosts, UN-funded cleanups abroad, and occasional one-off foreign sales.[^G-13^][^G-23^][^G-24^][^G-27^] Entirely absent: city utilities buying directly, small industrial producers, landfill operators buying equipment, and any pay-by-the-gallon commercial service. **Verdict: GA competes only for large government and national programs.** Its corporate structure makes it unlikely to chase the small, modular segment unless something external changes — and the specific trigger to watch is a Pentagon Success Memo for its Charlotte demo.[^G-13^]

---

3. Aquarden Technologies (product brand: SuperOx)

Bottom line: Aquarden owns the scarcest asset in this industry — industrial SCWO plants verifiably running for a real customer — and the weakest balance sheet. Its net worth is negative (−DKK 30.9 million at end-2025), it has lost money five years in a row (about DKK 49.5 million in total), it has 11 employees, and nearly all of its commercial life runs through one partner, ArianeGroup. The Danish EPA’s own conclusion — “the main obstacle

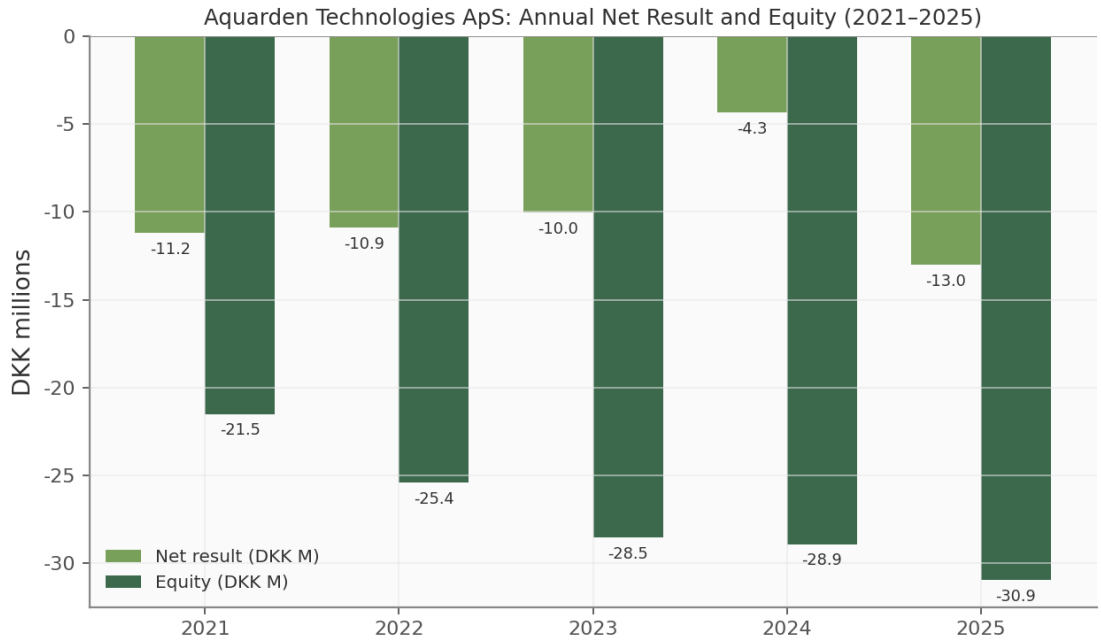
now for onsite implementation is the capacity and throughput" — is the competitor's size ceiling stated by its own regulator.

3.1 Financial health: family-owned, loss-making, worth less than nothing on paper

Aquarden Technologies ApS (founded 2005, based in Skævinge, Denmark) is owned by founder and sole director Zhuoyan Cai through a holding company; no outside or venture investors appear in the registry.[^A-2^][^A-3^] Danish registry accounts (figures in thousands of Danish kroner; 1 krone ≈ $0.15) show a company shrinking as it tries to commercialize:

DKK '000	2021	2022	2023	2024	2025
Gross profit (sales minus direct costs)	1,795	−532	1,111	5,843	**−1,528**
Net result (profit/loss after everything)	−11,177	−10,886	−10,029	−4,343	**−13,025**
Equity (what owners' stake is worth)	−21,505	−25,391	−28,520	−28,913	**−30,947**
Cash	4,639	4,467	1,198	1,213	2,706
Employees (average)	13	13	8	10	**11**

Source: Danish CVR/Virk annual reports via Proff.dk. Note: the company is exempt from publishing its revenue under Danish small-company rules; gross profit of DKK 1.1–5.8 million in its best years implies revenue of at most the low tens of millions of kroner (roughly €1–4 million) — our estimate, flagged as such.[^A-1^]



Aquarden net result and equity, 2021–2025. Source: Danish CVR registry annual reports via Proff.dk.[^A-1^]

Read the table this way: in its best year (2024) Aquarden's direct project margin was DKK 5.8 million — about $900,000 — which never came close to covering its overhead. In 2025 even the direct margin went negative, meaning projects cost more to deliver than customers paid, and the loss was the biggest on record.

2025 was the worst year yet: the largest loss (−DKK 13.0 million), negative gross margin, and cash of DKK 2.7 million against roughly DKK 7.3 million of annual payroll — a few months of runway unless the owner puts in more money.[^A-1^] The company is carried by about DKK 77 million of growing "other debt" plus about DKK 6.8 million owed to related parties — in plain terms, owner loans, not outside investment.[^A-1^] Product development has been financed by grants: a Danish EPA "PFAS-inator" project and €1.59 million of French state aid inside the €5.03 million ArianeGroup ELIXIR project.[^A-10^][^A-15^] Danish trade press called it a "deficit-hit cleantech company" back in 2021, and under Danish law a company with persistently negative equity needs owner support to keep operating.[^A-17^] **What this means for HCSP:** Aquarden cannot offer customer-friendly financing (no leasing, no "we build and run it for you"), cannot afford an international service network, and is one broken relationship away from crisis — which makes its technology credibility attractive to acquirers and its customers available to better-funded rivals.

3.2 Customers: two real industrial systems, one controlling partner

Aquarden's entire operating commercial fleet is two industrial SuperOx units, both at ArianeGroup sites in France — the first (2018) destroying rocket-propellant and explosives waste, the second (delivered 2024) destroying PFAS — plus Aquarden's own full-scale plant in Denmark, used for paid customer test runs.[^A-5^][^A-8^] Everything else in the public record is pilots and lab tests: Stockholm's Arlanda Airport (2020), a Swedish landfill's leachate (2019–20), a U.S. EPA-directed foam test run on Aquarden's **lab-scale** reactor in Denmark at just 1 liter per hour (2021) — in which Aquarden's leftover PFOS in the treated water was the highest of the three vendors tested — and a hospital-wastewater demonstration in Copenhagen.[^A-7^][^A-11^][^A-12^][^A-13^]

The strategically decisive fact is who owns the customer relationship. ArianeGroup now markets **ELIXIR as its own waste-destruction service** — "fixed or mobile facilities, direct purchase or leasing," per its 2026 datasheet, with the next PFAS facility planned for Toulouse — all built on Aquarden machines; selling outside France is delegated to small partners (PFAS Busters LLC in Japan; treatment-train partners in Denmark).[^A-7^][^A-9^][^A-15^][^A-19^][^A-27^] Aquarden has **no U.S. presence at all**: the EPA test ran in Denmark, the CEO's 2023 remark about being "in talks with U.S. companies" remains just talk, and the company is absent from every U.S. defense program.[^A-11^][^A-18^] **For HCSP, Aquarden is therefore not a U.S. competitor.** It is a European technology supplier whose shop window is being absorbed by a defense giant — a live demonstration of what happens to an underfunded SCWO vendor that bets everything on one channel partner.

3.3 How small can Aquarden go — and who it sells to

The SuperOx reactor handles about 5 tonnes per day (up to 12 cubic meters of liquid waste daily, roughly 180 liters per hour continuously), uses only electricity (80–300 kWh per cubic meter), and — critically — comes in only one size. Growing means buying more identical reactors.[^A-6^][^A-7^][^A-15^] Third-party estimates put machines of this class at $2–10 million each to buy, with destruction costs of roughly $100–500 per tonne of concentrated waste.[^A-21^] The decisive evidence is the Danish EPA's final report on the PFAS-inator project: SCWO works technically, but *"the main obstacle now for onsite implementation is the capacity and throughput"* — so it is positioned as the expensive last step of a treatment chain, after other equipment has concentrated the waste, not as a standalone plant.[^A-15^][^A-16^] An independent corporate guide reaches the same conclusion: SCWO "require[s] high capital investment and skilled operation, often limiting their early adoption to larger regional facilities or remediation contractors."[^A-20^]

Aquarden's real customers are big European defense/aerospace groups (essentially just ArianeGroup), plus — as targets — airports, fire services, military bases, and industrial sites with small volumes of very nasty, high-liability waste: old firefighting foam concentrate, spent filter media, explosives.[^A-7^][^A-8^][^A-29^] Cities and small sites are reached only indirectly, as waste producers shipping concentrate to someone else's plant. There is no named municipal SuperOx purchase and no small-format product.[^A-15^] The honest qualification: Aquarden's destruction performance (better than 99.99% on PFAS concentrates and explosives, at industrial scale, in France) is genuinely demonstrated and rare. Its strength is engineering credibility plus the ArianeGroup channel — not size, cost, or reach.[^A-5^][^A-15^] HCSP's counter is economic, not technical: serve the waste streams and site sizes that a 5-tonne-a-day, $2–10 million machine structurally cannot.

---

4. Revive Environmental (product: Battelle PFAS Annihilator)

Bottom line: Revive is the best-backed and best-permitted SCWO player in North America — and it has deliberately chosen a business model that skips small waste producers. You do not buy a Revive machine; you ship your waste to one of its two permitted plants (Columbus, Ohio and Grand Rapids, Michigan) and pay by the gallon. That choice leaves on-site and small-facility demand open by design. At the same time, Revive's third-party-verified "Certificate of Annihilation" — a lab-verified proof that a specific batch of waste was destroyed — has become the paperwork standard that HCSP will have to match.

4.1 Who stands behind it

Revive Environmental Technology, LLC was spun out of Battelle Memorial Institute (the giant nonprofit research organization) in December 2022 and launched publicly on January 17, 2023. It is jointly owned by Battelle and the hedge fund Viking Global Investors, “with each sharing governance.” No valuation, financial terms, or later funding rounds have been disclosed.[^R-1^] Battelle contributed the patented PFAS Annihilator (its SCWO technology) after a decade of in-house research and keeps strategic control — a Battelle executive vice president chairs Revive’s board.[^R-2^][^R-6^] Research firm BlueTech classifies Revive as private, seed-stage, early-revenue, with a pay-per-treatment service model.[^R-5^] Revenue is undisclosed; the identifiable pieces — $668,258 (New Hampshire), $3 million (Ohio Round 1), and a $16.6 million appropriation behind the New Jersey program — imply at most a low-to-mid tens-of-millions annual services business (our estimate, flagged).[^R-9^][^R-10^][^R-11^] Headcount is also undisclosed; a March 2025 plan to add 10–20 staff, described as significant growth, suggests roughly 30–70 employees (estimate).[^R-8^]

The caution flags are specific. Founding CEO David Trueba stepped down on March 6, 2026 — replaced by the chief commercial officer — a leadership change three years after launch, in the middle of scale-up.[^R-6^] Revenue is episodic and depends on government money: Ohio’s first round was paid from a one-time $3 million Monsanto legal settlement, not a recurring budget, and Ohio’s environmental agency is publicly “trying to find funds” to destroy what remains.[^R-10^][^R-19^] Growth means building more Annihilator units (five were counted at the Columbus site), winning state-by-state facility permits — Columbus only became Ohio’s first permitted PFAS-destruction facility in 2026 — and paying for waste-concentration equipment ahead of revenue, all carried by the two owners with no new outside money disclosed since launch.[^R-14^][^R-18^]

4.2 Customers: four state collection programs, one state lost, two fixed plants

Revive’s visible revenue is state-run firefighting-foam (“AFFF”) take-back programs: New Hampshire (first in the nation, July 2023; 10,000 gallons for $668,258 — about $67 per gallon all-in, including trucking to Michigan); Ohio (Round 1 finished August 2026: 13,500+ gallons from 120 fire departments and airports); North Carolina (a statewide contract won March 2025 in a competitive pilot against three rivals; Phase 1 completed January 2026 with independent lab verification); and New Jersey (contract announced March 2026; about 150,000 gallons from 400+ fire departments, behind a $16.6 million state appropriation).[^R-9^][^R-10^][^R-11^][^R-13^][^R-22^][^R-23^] **Colorado went the other way: it chose competitor Aquagga’s non-SCWO process at a $40-per-gallon program payment** — proof that SCWO can lose state business on price.[^R-24^] Beyond take-backs: the “4never” landfill-leachate consortium (with Heritage-Crystal Clean, Allonnia, and EPOC Enviro) destroys 300–500 gallons a day of concentrated leachate at a reported $0.15–0.40 per gallon of raw leachate; Pentagon demonstrations including an April 2025 field demo in Detroit; and airport foam-transition work marketed around the third-party Certificate of Annihilation verified to EPA lab Method 1633.[^R-16^][^R-25^][^R-28^][^R-29^][^R-35^] The Pentagon opportunity is still prospective: February 2026 Defense Department guidance lists SCWO among “emerging options… [to] monitor” while naming landfills and incineration as the established routes — so Revive is not yet a default military disposal pathway even with more than 2 million gallons of military foam needing disposal by the October 1, 2026 phase-out deadline.[^R-26^][^R-27^]

All destruction happens at two permitted fixed plants — Columbus, OH (five Annihilator units as of the EPA Administrator’s July 7, 2026 tour) and Grand Rapids, MI (inside a Heritage-Crystal Clean site, running since May

2023 as the first permitted commercial SCWO PFAS-destruction operation in the U.S.).[^R-14^][^R-16^][^R-17^] The units are containerized and were once described as mobile, but commercially the waste travels, not the machine: “We concentrate the PFAS on-site, then destroy the concentrate at our permitted facility.”[^R-34^] The 2025 WEFTEC award citation — “North America’s first fully permitted SCWO-based PFAS destruction system operating at commercial scale” — is both Revive’s core strength and its self-description as a facilities business, not an equipment seller.[^R-20^]

4.3 How small can Revive go — and who it sells to

Each Annihilator unit processes roughly 500 gallons a day of concentrated waste; the company claims up to 7 million gallons a year of total capacity across its fleet.[^R-16^][^R-37^][^R-38^] The physics of the model shut out small, scattered producers directly: waste must be trucked as concentrate — or concentrated on-site and then trucked — to Ohio or Michigan, and freight, tracking paperwork (“chain of custody”), and per-project overhead impose an effective minimum economic volume. Small producers get served only when a government gathers them together — Ohio’s collection events, for instance, cap each waste owner at under 1,000 gallons per drop.[^R-34^][^R-39^] Revive’s own words concede the orientation: treatment is “cost-effective at scale for states, fire departments, airports, and commercial customers.”[^R-14^] Independent evidence confirms the pattern industry-wide: participants in the Pentagon demos concluded that treating waste on-site beats trucking it on cost and logistics; the existing filter-media reactivation suppliers refuse site contracts below about 36,000 kg per batch; and Minnesota’s pollution agency found that small wastewater plants face per-pound PFAS costs more than six times those of large plants.[^R-36^][^R-40^][^R-43^]

So Revive’s customers are state agencies (as contract counterparties), landfill operators (via the consortium), airports, the Pentagon (prospectively), and — by company claim — Fortune 500 industrials, all paying service fees by the gallon, never buying equipment.[^R-28^][^R-31^] The unserved segments map directly onto HCSP’s opening: small and remote producers whose volumes cannot justify Midwest freight; small continuous producers of dilute waste (95% of U.S. public water systems serve under 10,000 customers and lack the staff and budget for treat-and-truck strategies); customers who want to own equipment rather than sign a chain-of-custody shipping arrangement; and sludge/biosolids producers, since Revive’s machines handle only liquids, slurries, and concentrates.[^R-37^][^R-40^][^R-44^] **One capability HCSP must treat as mandatory:** Revive’s third-party-verified Certificate of Annihilation has become the de facto paperwork requirement in state buying processes — North Carolina’s program explicitly required verified destruction — and any HCSP offer without an equivalent certificate will be disqualified before price is ever discussed.[^R-13^][^R-29^]

---

5. Customers and Where They Operate

This section answers two practical questions for each competitor: who actually pays them, and in which countries do their machines or services actually run? The answer, in every case, is the same shape: a short list of government or government-funded customers, in one region of the world, with no global channel for reaching small facilities. Each table below calls out the largest and smallest verified engagement. Where a headline number is easy to misread, we say so — three examples: New Jersey’s $16.6 million is a **state appropriation** (the state’s total budget for its program), not Revive’s contract value, which is undisclosed; Blue Grass’s $5.3 billion is the

joint-venture program total, not GA's share, which is undisclosed; and Toulouse's €5.03 million is **ArianeGroup's project budget**, not Aquarden's contract revenue.

5.1 374Water's customers

Customer / site	Location	Type	Value	Status (late 2026)
City of Olathe — **LARGEST**	Olathe, KS	City utility (equipment sale via contractor Garney)	**$4,880,000 fixed price** ($3.0M for the machine; $2.2M received so far)	Signed, in fabrication — largest equipment order in company history [^W-c15^][^W-c16^][^W-c17^]
Orange County Sanitation District	Fountain Valley, CA	City utility (equipment sale, 2022)	~$4.6M to 374Water (district's all-in budget $8.7M)	~4 years late; factory test passed June 2026; delivery planned October 2026, not confirmed completed [^W-c9^][^W-c11^] [^W-c15^]
City of Orlando (Iron Bridge)	Orlando, FL	City utility (374Water runs the machine under license)	Demo $812K (2024); 5-year license (2026); company projects $3–5M/yr, up to $10–14M/yr expanded	**Operating** — flagship hub since March 2026 [^W-c1^][^W-c2^] [^W-c3^]
State of North Carolina / UNC	Statewide NC (processed in Orlando, FL)	State foam take-back program	~$1M initial; company says later phases "could be worth north of $5,000,000" — potential, not booked	Phase 1 (1,000 gal) begun November 2025 [^W-c29^][^W-c30^] [^W-c31^]
Pentagon demo at Clean Earth	Detroit, MI	Federal demo hosted at a hazardous-waste facility	Undisclosed	Completed 2025 (900 gal, >99.9% destruction; Pentagon "Success Memo" August 2026) [^W-c23^][^W-c26^][^W-c27^]
Pentagon demo at Peterson Space Force Base	Colorado Springs, CO	Federal demo + university	Undisclosed	Completed 2025 [^W-c28^]
City of St. Cloud — **SMALLEST** (with a disclosed figure)	St. Cloud, MN	City utility (crewed mobile pilot, state grant-funded)	**$600,000**	Completed September 15, 2026 [^W-c19^] [^W-c20^]
Crystal Clean (host site)	Oregon, OH	Hazardous-waste company partner	Undisclosed; no revenue yet	Announced September 2025; nothing built [^W-c32^]

A precision note on the ranking: the federal demos (Detroit, Peterson, plus a 2022 Navy award and a 2026 Air Force subcontract) carry **no disclosed values**, so they cannot be ranked on dollars.[^W-c34^][^W-c18^] The largest disclosed contract (Olathe, $4.88M) is about eight times the smallest disclosed one (St. Cloud, $600K) — a narrow pilot-to-first-equipment-sale band, with no booked contract above $5 million to date.[^W-c15^][^W-c20^]

Where 374Water actually operates: the United States only, across 8 states. Florida (the operating Orlando hub plus manufacturing), California (Orange County, in delivery), Kansas (Olathe, in fabrication), Minnesota (St. Cloud, completed), Michigan (Detroit demo, completed), Colorado (Peterson demo, completed), North Carolina (headquarters plus the state foam program), and Ohio (an announced Crystal Clean site, unbuilt); Indiana was a historical fabrication site, now vacated.[^W-c3^][^W-c6^] **There are no international sales, deployments, or revenue.** Australia, Europe, and Asia appear only as ambitions — a CEO interview in October 2025 and the "third horizon" of the September 2026 strategy announcement — with no country, partner, or timeline attached.[^W-c36^][^W-c37^] The only non-U.S. footprint is a secondary stock listing in Frankfurt — a trading venue, not an operation.[^W-c25^]

5.2 General Atomics' customers and hosts

Customer / site	Location	Type	Program size	Status (late 2026)
Blue Grass Army Depot / Bechtel-Parsons team — **LARGEST by program value**	Richmond, KY	Army chemical-weapons program (GA was the SCWO subcontractor)	**$5.3B joint-venture program total** ($2B+ built by 2023, including a dedicated SCWO building); **GA's own share is undisclosed** — do not read $5.3B as GA revenue	Built but **rejected August 2020**; waste trucked to Veolia, Texas [^G-c2^][^G-c5^][^G-c6^]
UN/GEF Central Asia DDT project	Kyrgyzstan (Suzak sites) & Tajikistan (Vakhsh and others)	Foreign governments via UN aid program	**$44.18M total** ($15.12M grant + $29.06M co-financing); 5,000-tonne DDT target; runs through December 2027	In implementation; GA's transportable unit is the named destruction technology [^G-c25^][^G-c28^] [^G-c30^]
U.S. Army Corps of Engineers demo via Bay West	Lake Elmo, MN	Federally funded demo through an environmental contractor	Bay West's prime contract is **$9,999,105** (GA was subcontracted)	Completed May–June 2025; follow-on work into 2026 [^G-c18^] [^G-c19^][^G-c20^]
Pentagon demo at Clean Earth	Charlotte, NC	Federal demo hosted at a hazardous-waste facility	Undisclosed	Completed August–September 2025; no results report or Success Memo located [^G-c12^][^G-c14^]

Tianjin Saipeng (largest unit sale)	Tianjin, China	Foreign commercial hazardous-waste operator	Three systems, sold April 2018; value undisclosed	**Commissioning never publicly confirmed** [^G-c23^]
Explo Systems at Camp Minden — **smallest named deployment**	Camp Minden, LA	Private explosives recycler on an Army contract	One 3 gpm unit; limited testing only	**Collapsed** when the customer went bankrupt; unit stranded [^G-c8^][^G-c9^]
Tooele Army Depot	Tooele, UT	Army disposal depot	Undisclosed (small add-on unit)	**Abandoned** — EPA cites capacity and reliability problems [^G-c1^]
Kurita Water / Komatsu (historical)	Japan	Industrial joint-venture demo	2 gpm demo unit, 1,000+ operating hours	Completed, early 2000s [^G-c33^]
GA's own San Diego test facility + AxNano test work	San Diego, CA / North Carolina	Vendor-owned test site; small paying test customer	>6M gallons tested cumulatively	Operating [^G-c34^][^G-c36^]

Where GA actually operates: 6 countries — and every current U.S. project is government-funded. The realized footprint is the U.S. (Utah, Kentucky, Louisiana, Oklahoma, North Carolina, Minnesota, California), Japan (a historical demo), China (the 2018 Tianjin sale), and Kyrgyzstan and Tajikistan (the UN-funded DDT project running through 2027).[^G-c8^][^G-c23^][^G-c30^][^G-c33^] Everything currently active in the U.S. is paid for by the Pentagon or the Army Corps of Engineers; everything abroad runs through UN/multilateral aid programs or one-off foreign sales. GA's corporate empire has offices in London, Abu Dhabi, New Delhi, Tokyo, and Switzerland, and products in 68 countries — but **there is no PERSES-dedicated overseas sales office, distributor, or partner**, and no standing commercial channel for the waste line.[^G-c38^] Announced but unrealized: additional Minnesota and Georgia installations (October 2025).[^G-c14^]

5.3 Aquarden's customers and test hosts

Customer / site	Location	Type	Value	Status
ArianeGroup ELIXIR Unit 2 (PFAS) — **LARGEST documented engagement**	Toulouse, France	Defense/aerospace prime; commercial PFAS-destruction hub	**€5.03M total project, incl. €1.59M French state aid** — caveat: this is **ArianeGroup's project budget**, not Aquarden's contract revenue (undisclosed; machines of this class are est. $2–10M)	Delivered June 2024; permitted summer 2025; operating [^A-c3^][^A-c4^][^A-c5^] [^A-c6^]
ArianeGroup ELIXIR Unit 1 (energetics)	Saint-Médard-en-Jalles (near Bordeaux), France	Defense/aerospace; destroys propellant and explosives waste from named suppliers (ROXEL, EURENCO, SNPE)	Undisclosed; 100–200 tonnes/yr per reactor	**Operating since 2018** [^A-c1^][^A-c2^]

French fire services SDIS 16 & 17	Charente / Charente-Maritime, France	Municipal fire services — **indirectly**, as customers of ArianeGroup's ELIXIR service	Undisclosed (state grant-funded); 23 fire trucks decontaminated, rinse waste destroyed at Toulouse	Announced / in execution (2025–26) [^A-c10^]
Danish EPA "PFAS-inator" project	Denmark (11-partner consortium)	Government research grant	DKK 4.85M total budget, DKK 2.8M grant (~€650K / ~€375K)	Completed December 2023 [^A-c16^][^A-c17^]
Stockholm Arlanda Airport / Swedavia	Stockholm, Sweden	State-owned airport	Undisclosed (test scale)	Test campaign 2020; **no unit installed**; the 2025 full-scale follow-on contract went to a filter-technology rival (Mellifiq) [^A-c11^][^A-c12^][^A-c13^]
Sörab (municipal waste company)	Northern Stockholm region, Sweden	Municipal landfill leachate	Undisclosed (samples shipped to Denmark)	Completed test 2019–20; PFAS cut from 3.7 to <0.035 µg/L [^A-c11^][^A-c12^]
U.S. EPA-directed foam test — **SMALLEST documented engagement**	Run on Aquarden's lab reactor in Skævinge, Denmark	U.S. government research program	Undisclosed; clearly small lab-bench scale (1 L/h, a single effluent sample)	Completed 2021; Aquarden's leftover PFOS was the highest of the three vendors tested; **no U.S. sale followed** [^A-c15^]
Royal Copenhagen Hospital	Copenhagen, Denmark	Hospital wastewater (60+ pharmaceuticals)	Undisclosed	Demonstration reference only [^A-c20^]
PFAS Busters LLC	Japan (Tokyo debut January 2026)	Market-entry partner	None	Announced introduction partner only; no installation, no end customer [^A-c21^]

Where Aquarden actually operates: verifiable industrial systems in exactly two countries — France and Denmark — and the Danish one is its own test plant, not a customer site. Both French units are owned and run by ArianeGroup, which sells the destruction service under its own ELIXIR brand. Sweden and Norway are test-campaign territory only. The U.S. is "in talks" (the CEO's December 2023 words) with no contracts.[^A-c3^][^A-c23^] Japan is a 2026 announcement with nothing behind it yet. The partner's claim that SuperOx is "deployed in more than five countries in Europe," and the trade-show listing of "Denmark, Singapore," are marketing statements with no named installation behind them.[^A-c21^][^A-c24^] The Danish EPA itself confirms there is **no commercial SCWO plant in Denmark receiving PFAS waste** — Aquarden's home plant serves test campaigns.[^A-c19^]

5.4 Revive's customers

Customer / program	Location	Type	Value	Status (late 2026)
New Jersey foam program — **LARGEST program**	NJ statewide; waste trucked to Columbus, OH	State environmental agencies; logistics by Republic Services	**$16.6M state appropriation** — caveat: that is the	

			state's program budget, not Revive's contract value, which is undisclosed; ~150,000 gal from 400+ fire departments	In execution; >150,000 gal collected and trucked to Ohio by July 2026 [^R-c11^][^R-c12^][^R-c13^]
Ohio Round 2	Ohio statewide	State environmental agency	**>$13M** directed to the program from a DuPont settlement	Registration closed September 2026; collection from November 2026 [^R-c7^]
Ohio Round 1 — **largest fully documented award**	Ohio statewide; destruction in Columbus	State environmental agency (awarded to Battelle, Revive subcontracted)	**$3M** from a Monsanto settlement; 13,500+ gal from 118 fire departments + 2 airports	Completed August 2026 [^R-c4^][^R-c5^]
North Carolina foam take-back	NC statewide; destruction in Columbus, OH	State-academic program	Undisclosed; Phase 1 = 1,000 gal, expandable to ~60,000	Phase 1 completed and independently verified January 2026 — **first provider to finish** [^R-c9^][^R-c10^]
New Hampshire — **SMALLEST documented contract**	NH statewide (10 county collection events); destruction in Columbus, OH	State environmental agency	**$668,258** for ~10,000–12,000 gal (~$67/gal all-in)	Completed August 2024 [^R-c1^][^R-c2^][^R-c3^]
Colorado foam take-back	Colorado statewide	State environmental agency	$40/gal participant payment; 3-year, ~40,000 gal	**LOST to competitor Aquagga** (~May 2026) [^R-c14^][^R-c15^]
4never landfill-leachate consortium	Hub: Grand Rapids/Wyoming, MI; feedstock trucked/railed from multiple states (3+ unnamed Michigan landfills)	Private landfill operators (contract via Heritage-Crystal Clean)	Fixed price per gallon; $0.15–0.40/gal of raw leachate	Operating since May 2023; expanded November 2024 [^R-c17^][^R-c18^][^R-c19^]
Pentagon demos	Detroit, MI (done 2025); Wurtsmith AFB, MI (planned, unconfirmed)	Federal (defense research programs)	Undisclosed	Demo executed; still in reporting; not yet a default Pentagon pathway [^R-c22^][^R-c23^]
Fort Wayne International Airport	Fort Wayne, IN	Airport (fire-truck foam transition; ~950 gal)	Undisclosed	Completed late 2024 [^R-c24^][^R-c25^]

Where Revive actually operates: the United States only, and all destruction happens at two Midwest sites. The two permitted plants are Columbus, Ohio (five Annihilator units, the headquarters, permitted in 2026 as Ohio's first PFAS-destruction facility) and Grand Rapids/Wyoming, Michigan (inside a Heritage-Crystal Clean

site, running since May 2023).[^R-c17^][^R-c27^] Contracted collection programs span four states (Ohio, New Hampshire, North Carolina, New Jersey), plus airport work in Indiana and Pentagon demos in Michigan — but in every case the waste travels to the Midwest by truck or rail; the machines do not travel to the waste.[^R-c13^][^R-c19^] The 4never consortium is marketed as available in "the United States and parts of Canada," but **no documented Canadian or any other non-U.S. customer, deployment, or permit exists**.[^R-c28^] Colorado — the only state program west of the Mississippi that Revive contested — went to a rival.[^R-c14^] A west-coast expansion of its separate filter-media regeneration business is planned only for 2027.[^R-c30^]

5.5 What the map shows

Put the four maps together and the pattern is stark. **374Water is US-only (8 states, no international revenue). GA spans 6 countries but every current project is government- or aid-funded. Aquarden has exactly two verified industrial systems, both in France, plus its own Danish test plant. Revive is US-only, with all destruction at two Midwest addresses.** No competitor has a global channel, a distributor network, or any mechanism for reaching a small facility in, say, Indonesia, the Caribbean, or rural America. Every competitor is regional and government-heavy — which means HCSP's first customers will be buyers none of these four companies can reach, quote, or service, in markets where none of them has a footprint.

---

6. The Scale Gap — Why Small Facilities Are Left Out, and What HCSP Should Do About It

6.1 The four competitors side by side

Question	374Water (AirSCWO)	General Atomics (PERSES)	Aquarden (SuperOx)	Revive (Annihilator)
Financial health	Auditors warn it may run out of money; $1.78M cash; $57.2M in piled-up losses [^W-3^][^W-6^]	Parent earns ~$3.2B/yr; waste line is a tiny side business [^G-3^][^G-4^]	Worth less than nothing on paper (−DKK 30.9M); losses every year 2021–25; 11 staff [^A-1^]	Backed by Battelle + Viking Global; finances undisclosed; CEO replaced March 2026 [^R-1^][^R-5^] [^R-6^]
Smallest offering	AS1 (1 tonne/day, mobile) — **kept for its own crews, not sold** [^W-4^]	3 gpm container (~4,300 gal/day) plus support equipment [^G-9^]	One 5 tonne/day reactor; grow only by adding identical units [^A-6^]	~500 gal/day units at two central plants [^R-16^]
Smallest deal we could verify	$4.88M equipment project (Olathe); $600K crewed pilots [^W-6^][^W-24^]	No priced deals at all; demos are government-funded [^G-13^][^G-27^]	Machines est. $2–10M each (third-party estimate) [^A-21^]	Pay-per-gallon service; small volumes only via state collection events (capped under 1,000 gal) [^R-39^]
Who they sell to	City utilities, state programs, the			

	Pentagon [^W-5^] [^W-28^]	Pentagon/Army programs, waste-facility hosts, UN-funded cleanups [^G-13^] [^G-23^]	One European defense prime (ArianeGroup); partner treatment chains [^A-8^][^A-15^]	State agencies, landfill consortium, airports, Pentagon demos [^R-9^][^R-28^]
Who they do NOT serve	Buyers under $1M, private industry, small/rural sites [^W-4^][^W-32^]	Anything under 3 gpm; every kind of commercial service [^G-9^][^G-16^]	Small facilities, bulky waste streams, the entire U.S. market [^A-15^][^A-18^]	Small/remote/continuous producers; equipment buyers; sludge [^R-37^][^R-44^]

The consistency of this table is the finding. Four competitors with four completely different ownership structures — a small public company, a private defense giant, a family-owned Danish firm, and a nonprofit/hedge-fund joint venture — have independently arrived at the same excluded customer: the small facility that wants waste destroyed on its own site for under $1 million. That is not coincidence. It reflects the shared economics of the technology class as it exists today — expensive high-pressure reactors, the need for skilled operators, permitting costs, and the fact that most of the cost in industry projections of $12–90 per tonne is paying off the equipment itself.[^G-31^] Even the closest non-SCWO rival fits the pattern: Aquagga's HALT process won Colorado at $40 per gallon as a centralized program service, not as a small on-site product.[^R-24^] The gap is a fact about how this market is structured, not a hole in any one company's catalog.

6.2 The evidence for the gap

Four independent "floors" — the lowest each competitor can go — bracket the unserved space. **Purchase-cost floor:** the cheapest equipment transaction we could verify anywhere is 374Water's $4.88M Olathe project, and its smallest machine is not sold at any price.[^W-4^][^W-6^] **Size floor:** GA's 3 gpm baseline (~4,300 gallons a day) is a full industrial plant needing a federally permitted host site or a government sponsor; nothing smaller exists.[^G-9^][^G-13^] **Machine-class floor:** Aquarden's single 5 tonne/day reactor, in the $2–10M range, is the minimum economic unit — and its own regulator names capacity and throughput as the obstacle to on-site use.[^A-15^][^A-21^] **Shipping floor:** Revive's model prices small producers out through freight and paperwork unless a state collects their waste for them — and even then caps each drop-off below 1,000 gallons.[^R-34^][^R-39^]

The demand side of the gap is documented by regulators and by the competitors' own ecosystem: 95% of U.S. public water systems serve fewer than 10,000 customers; small plants pay more than six times as much per pound of PFAS as large ones; and participants in the Pentagon's own demos concluded that treating waste on-site beats trucking it "because of the logistical and permitting challenges of moving waste from one place to another."[^R-40^][^R-43^][^R-44^] **No credible sub-$1M, on-site, small-facility SCWO offer exists.** That is the segment HCSP's modular design — with its self-fueling hydrogen loop and its saleable co-products (distilled water, power, cooling) — is built for. The co-products matter precisely because they can make a small site pay for itself where a competitor's pure disposal-cost math cannot.

6.3 What HCSP must do to hold the opening

The opening is real but time-limited. Holding it takes five concrete moves, each tied to a documented competitor failure or capability:

1. **Ship small, customized models — and say so.** Every competitor's floor is set by one industrial machine size. HCSP's product line should be explicitly sized below the $4.9M purchase-cost floor and below 3 gpm, offered in versions matched to each industry, with customization stated as a capability. That product definition *is* the segment definition.[^W-6^][^G-9^]
2. **Offer on-site destruction priced per gallon, as a service option.** Revive's benchmarks — $0.15–0.40 per gallon of raw leachate and about $67 per gallon all-in for firefighting foam — set the price ceiling. An on-site HCSP service eliminates the freight line item that Pentagon demo participants identified as decisive. [^R-9^][^R-35^][^R-40^]
3. **Issue third-party-verified destruction certificates from day one.** Revive's Certificate of Annihilation (verified to EPA lab Method 1633) is now the de facto requirement in state tenders; HCSP must match it or be excluded before price is discussed.[^R-13^][^R-29^]
4. **Document reliability relentlessly, to escape the industry's graveyard reputation.** GA's abandoned Tooele unit and rejected Blue Grass system, plus 374Water's four-year-late, money-losing Orange County delivery, have primed buyers to expect SCWO projects to fail. Published run-time hours, uptime statistics, and reference-site performance are competitive weapons, not marketing garnish.[^G-19^][^G-20^][^W-2^]
5. **Assign a named owner to each of four watch items:** (a) a **Pentagon Success Memo for GA's Charlotte demo** — 374Water got its memo in August 2026, and a GA memo would unlock large defense orders; (b) **ArianeGroup's ELIXIR expansion** (Toulouse is planned) — a well-funded European service competitor built on Aquarden's machines; (c) a **Revive pivot to on-site deployment** — its containerized units already ran on-site at the Detroit demo, so the move is technically available to it; (d) **Aquagga's price competition** — Colorado's $40/gal award shows states will pick non-SCWO destruction on price.[^W-28^][^G-13^] [^A-27^][^R-24^][^R-25^]

The common thread: HCSP's window is defined by the competitors' structural limits — money, organization, permits, freight — not by their ignorance of the small-facility segment. Windows defined by constraints close when the constraints lift: a 374Water refinancing, a big GA defense contract, a Revive on-site pivot. **Speed to working reference installations matters more than a long list of leads.**

---

7. Money That Doesn't Cost Equity: Grants, Prizes, and Partner Channels

Bottom line: beyond the existing 80-program U.S. register, we screened 35 additional channels for funding that does not require selling shares — "non-dilutive" money: grants, prizes, pilot funding, and minority investments. **Eight are open to HCSP directly today; nineteen are reachable only through a partner organization that officially receives the money; six are ruled out.** The best additions are the Asian Development Bank's Technology Innovation Challenge (the cleanest true grant open to a for-profit SCWO vendor), the AB InBev 100+ Accelerator (pilot cash plus a door into five big beverage companies), WaterEquity's Water & Climate Resilience Fund ($2–15M minority investments), and the Water-AI Nexus at the University of Pennsylvania (the strongest no-equity entry into the data-center water market).

7.1 Channels HCSP can apply to directly

#	Channel	What it is	How much	Can HCSP apply?	Timing	Best-fit market
1	Asian Development Bank Technology Innovation Challenge [^N-15^]	Pilot grant	Up to $450K (you cover 10%)	**Yes, directly** — open to tech companies piloting in ADB developing member countries	Themed calls; watch challenges.adb.org	Islands/Pacific; industrial wastewater — **top pick**
2	AB InBev 100+ Accelerator (with Coca-Cola, Colgate-Palmolive, Danone, Mondelēz, Unilever) [^N-34^][^N-35^]	Pilot funding + accelerator program	Up to $100K per pilot, 6-month program	**Yes** — built for for-profit startups; water stewardship is a standing theme	Annual cohort (~spring)	Distillery/brewery wastewater + water reuse
3	WaterEquity Water & Climate Resilience Fund [^N-26^][^N-27^]	Investment fund taking minority stakes	$2–15M per deal; >$100M raised of a $200M target (Microsoft is the anchor investor)	**Yes** — open application for water companies in 8 emerging markets	Rolling	Commercial/industrial wastewater and reuse projects
4	Global Environment Facility (GEF) Challenge Program for Adaptation Innovation [^N-4^][^N-5^]	Competitive grant	~$10M per call; >$40M committed across 32 projects	**Yes** — the one GEF window that invites private-sector innovators; projects must deliver climate-adaptation benefits in developing countries	Periodic calls (funds flowing through 2026)	Islands/remote climate-resilient water
5	Water Center at Penn — Water-AI Nexus (with Amazon Web					

	Services, the Water Environment Federation, and Leading Utilities of the World) [^N-60^]	Research consortium / shared-cost pilots	Negotiated per project	**Yes** — industry collaborators join and co-fund pilots	Rolling	Data-center cooling water — **top pick for that market**
6	WEF/WRF "LIFT" Technology Scans [^N-61^] [^N-64^]	A vetting program run by ~300 utilities (no cash, but credibility and adoption risk-sharing)	In-kind	**Yes** — designed for technology vendors	Rolling/annual	Biosolids, PFAS, industrial wastewater
7	Breakthrough Energy Discovery — Explorer grants [^N-54^][^N-55^]	Philanthropic grant	$50–250K (very early-stage); a Fellows track up to $500K as a convertible instrument (that part does dilute)	**Yes, but fit is a stretch** — applicants must show a path to cutting 500M tonnes of CO_2 a year at scale; only viable via a biosolids-methane-plus-clean-power framing	Annual; next concept-paper deadline reported November 2, 2026	Biosolids/energy only
8	ClimateWorks targeted requests for proposals [^N-52^][^N-53^]	Grant (per specific call)	Pool ~$9M (2025 Adaptation & Resilience Fund)	**Sometimes** — past calls explicitly allowed private companies; no standing application	Episodic; watch the newsletter	Water resilience/adaptation

Read this table as a portfolio, not a menu. The ADB and GEF windows pay for exactly the story HCSP can tell and pure-disposal competitors cannot: in island and remote settings, HCSP's distilled-water and power co-products are a climate-resilience pitch, not just waste destruction. Both require deploying in a developing country, so they pull HCSP's international pilot plans forward. The 100+ Accelerator and the Water-AI Nexus are door-openers: the pilot money matters less than sitting across the table from five beverage giants, or alongside Amazon Web Services and the Water Environment Federation. WaterEquity is patient investment capital, not a grant — engage it once an emerging-market project company exists (its WaterConnect sister program helps prepare projects that are not yet big enough to fund).[^N-26^] Breakthrough Energy carries an honest caveat: the 500-million-tonne CO_2 test is a genuine fit risk for a water/PFAS company, so do not spend proposal effort until the biosolids-methane math is credible. LIFT costs nothing but time and buys the utility credibility that the reliability strategy in Section 6.3 needs. Two further channels were assessed and deliberately deprioritized rather than padded in: Xylem's Partnerships Accelerator is invitation-only and unfunded (its value is a sales channel; it requires a proven, market-ready product with a paying customer — a 2027 item), and UNICEF/UN purchasing is a weak fit for hardware of SCWO's size.[^N-44^][^N-24^]

7.2 Channels reachable through a partner

The nineteen partner-dependent channels share one mechanic: the money is paid to a nonprofit, an accredited institution, a utility, or a government borrower, and HCSP participates as the technology supplier, joint-venture partner, or project company. They fall into four families:

- **Climate funds via accredited institutions.** The Green Climate Fund finances projects (its Simplified Approval Process goes up to $25M per project) but only organizations it has accredited can apply; HCSP would join as the technology partner on, for example, a wastewater-and-reuse project for a small island state. The dedicated GCF small-business pilot window is closed — watch only for a successor.[^N-1^][^N-2^] The GEF's Non-Grant Instrument (up to $15M per project; a $196M blended-finance envelope in the current funding round) flows through partner agencies such as the World Bank, IFC, ADB, and UNIDO.[^N-6^][^N-7^]
- **Development banks via their borrowers.** The World Bank's 2030 Water Resources Group assembles mixed public-private water projects in active country programs (an industrial water-reuse alliance in Bangladesh, plus Kenya, South Africa, and Colombia) and has facilitated $1.6B in water financing; IFC finances private water clients at scale; IDB Invest and IDB Lab cover Latin America and the Caribbean (industrial wastewater, islands, rum distilleries); the African Development Bank is a GEF implementing agency with >$400M secured; and the Caribbean Development Bank is the natural buying channel for the islands market.[^N-10^][^N-13^][^N-16^][^N-18^][^N-19^][^N-20^]
- **Corporate foundations via nonprofit implementers.** The Coca-Cola Foundation and the Grundfos (Poul Due Jensen) Foundation fund water nonprofits — for example, their joint $1.1M 2025 grant to the Global Water Center for East Africa — so the route in is partnering with an implementing nonprofit (Water Mission or Global Water Center types) where HCSP's distilled-water output is the deliverable.[^N-31^][^N-33^][^N-46^] PepsiCo, Nestlé, Diageo, Suntory, and Danone offer no vendor grant windows; engagement there is commercial or via the 2030 WRG platform — recorded honestly as sales channels, not grants.[^N-12^][^N-38^][^N-40^]
- **Research and community hosts.** EREF research grants (up to ~$300K; "emerging contaminants in leachate" is an explicit 2024–25 priority) go to university researchers — HCSP would co-propose an SCWO leachate/PFAS demonstration as site host and co-funder. Rotary Foundation Global Grants (budgets of $30K+, with $15–200K of matching funds, rolling) require local and international Rotary clubs as sponsors; useful for a small island or community demonstration and visibility, not core funding. UNEP-GEF chemicals projects (for example an €8.8M+ textile-sector program in Bangladesh, Indonesia, Pakistan, and Viet Nam) admit vendors as technology providers inside the project countries.[^N-65^][^N-66^][^N-22^]

Sequencing implication: partner channels are slow — 12 to 36 months from first conversation to funded project — but large. Seed them now, in parallel with the direct channels, not after them. Priority pairings: the Green Climate Fund and Caribbean Development Bank for the islands market; the 2030 WRG Bangladesh platform for textile-industrial wastewater; and EREF for landfill leachate. Each maps to a market where none of the four competitors (Sections 1–5) has any presence.

7.3 Ruled out — so no one wastes proposal effort

Six screened channels should receive no further work: **charity: water** grants only to vetted local nonprofits and its wells-and-filters portfolio is a technology mismatch;[^N-29^] **WaterAid** is a fundraiser, not a grantmaker — corporate money flows *to* it;[^N-30^] **IRC (WASH)** is a knowledge nonprofit with no vendor grant program; the **KR Foundation** explicitly excludes for-profit companies and funds advocacy, not market-ready technology;[^N-50^][^N-51^] the **Quadrature Climate Foundation** is invitation-only and funds nonprofits;[^N-48^] and the **Green Climate Fund's small-business pilot window is closed**.[^N-2^] The Ecolab Foundation and Xylem Watermark likewise restrict grants to nonprofits ($5–50K and partner-grant models respectively); those companies' real water money flows through WaterEquity and commercial partnerships instead.[^N-41^][^N-43^]

7.4 How this stacks onto the existing U.S. register

These 35 channels add to — they do not replace — the existing U.S. federal/state/municipal/private register (defense research programs, state PFAS funds, Imagine H2O, and similar). The recommended sequence runs three tracks in parallel. **Track 1 (now, 0–6 months):** submit to the ADB Technology Innovation Challenge and the next 100+ Accelerator cohort; start WaterEquity screening if any emerging-market project vehicle exists; enter a LIFT Technology Scan. Roughly $550K–$15M of money is at stake for proposal effort measured in weeks. **Track 2 (seed now, harvest in 12–36 months):** open accredited-partner conversations for Green Climate Fund island projects, join 2030 WRG country platforms where textile-industrial wastewater is live, and co-develop an EREF leachate proposal with a university lead researcher — these build the international reference sites that Section 6.3 says HCSP needs before competitor constraints lift. **Track 3 (conditional):** Breakthrough Energy only if the biosolids-methane case can clear the 500-million-tonne CO_2 bar; ClimateWorks calls as they appear; Xylem's accelerator once HCSP has a proven product and a paying, non-grant customer. The logic across all three tracks mirrors the competitive picture: every selected channel monetizes something — modularity, water and energy co-products, small-site economics — that 374Water, General Atomics, Aquarden, and Revive structurally cannot offer. Grant capital then compounds the segmentation advantage instead of merely extending runway.

The Words We Use (One-Page Glossary)

Term	What it means in this report
SCWO (supercritical water oxidation)	Destroying waste with very hot, high-pressure water (about 650 °C, 4,000 psi). Under those conditions organic waste chemically burns up inside the water. HCSP's core technology class.
PFAS	"Forever chemicals" — industrial chemicals (used in firefighting foam, nonstick coatings, and more) that barely break down in nature. Destroying them is the main demand driver.
AFFF	Aqueous film-forming foam — the PFAS-containing firefighting foam now being phased out and collected for destruction.

Going concern	An auditor's formal warning that a company may run out of money within a year. Both of 374Water's latest filings carry one.
10-K / 10-Q / 8-K	The annual, quarterly, and news-event reports U.S. public companies must file with the SEC. The most reliable source on 374Water's finances.
CapEx / purchase cost	The up-front price of buying and installing equipment (as opposed to "OpEx," the ongoing running cost).
Equity (negative)	What the owners' stake is worth on paper: assets minus debts. Negative equity means debts exceed assets.
ATM facility / baby-shelf rule	A program for selling new shares into the market bit by bit; SEC "baby-shelf" rules cap how much a very small company can actually sell this way.
Convertible note / warrant	A loan that can turn into shares (note), plus a coupon giving the lender the right to buy shares cheaply later (warrant) — expensive emergency money.
Reverse split	Swapping, say, 10 old shares for 1 new one to lift the share price and keep a stock-exchange listing. Cosmetic; signals distress.
MGD / gpm / tonnes per day	Size measures for water plants and machines: millions of gallons per day; gallons per minute; metric tonnes of waste processed per day.
WDS	374Water's "Waste Destruction Services" model — 374Water owns and crews the machine on the customer's site and charges for the service.
TSDF / RCRA permit	A treatment, storage, and disposal facility — a site with a federal hazardous-waste permit (under the Resource Conservation and Recovery Act). Hosting or shipping waste requires one.
DIU / ESTCP / Success Memo	Pentagon innovation programs that pay for technology demonstrations. A DIU "Success Memo" is the stamp of approval that turns a demo into a path for large orders.
Certificate of Annihilation	Revive's third-party-lab-verified proof that a specific batch of waste was destroyed (tested to EPA Method 1633). Now the standard paperwork in state tenders.
Non-dilutive funding	Money that does not require selling shares: grants, prizes, pilot funding, concessional loans.
Accredited entity	An organization officially approved (e.g., by the Green Climate Fund) to receive and manage climate funds; companies like HCSP participate through one.
Treatment train	A chain of treatment steps — e.g., first concentrate the waste with filters or foam fractionation, then destroy the concentrate with SCWO.
SAFF / GAC / IX	Concentration technologies that capture PFAS from large volumes of water into a small volume (foam fractionation; granular activated carbon; ion-exchange resin). SCWO then destroys the concentrate.
Take-back program	A state-run collection of old firefighting foam from fire departments and airports, paid for by the state.
DKK	Danish kroner; 1 krone ≈ $0.15. Aquarden's accounts are reported in DKK thousands.
HCSP	The modular SCWO wastewater system that is the subject of this report — designed for small and mid-size facilities, with saleable co-products (distilled water, power, cooling).

References

Companion reference file for `HCSP_Competitor_Intelligence_Plain.md` (the chapter itself contains no reference list). Citation indices are prefixed by source research file: `W` = comp_374water.md, `G` = comp_general_atomics.md, `A` = comp_aquarden.md, `R` = comp_revive.md, `N` = grants_ngo.md; `-c` indices (`W-c1` etc.) come from the customer-and-geography files (cust_374water.md, cust_ga.md, cust_aquarden.md, cust_revive.md) using each file's own numbering. Entries are copied from the reference lists of those verified research files (September–December 2026 research); only sources actually cited in the chapter are listed. No new supplementary (S-series) sources were added — every figure is preserved from the verified research.

374Water Inc. (AirSCWO) — from research/comp_374water.md

[^W-1^]: Quartz/SEC summary of FY2024 10-K (filed 2025-03-28) — https://qz.com/374water-inc-scwo-reports-earnings-1851773002 (Mar 28, 2025). "revenues of $445,445… net loss of $12,434,114… working capital of $11,760,131 and an accumulated deficit of $28,387,618… raised approximately $11.4 million through a registered direct offering."

[^W-2^]: 374Water FY2025 Form 10-K via last10k (SEC, filed 2026-03-31) — https://last10k.com/sec-filings/scwo. "revenue totaled $0.2 million… reduction in equipment revenue of approximately $1.9 million [OC San]… Net loss… $21.0 million… Cash… $3.2 million."

[^W-3^]: 374Water FY2025 Form 10-K, SEC EDGAR (accession 0001654954-26-003094, filed 2026-03-31) — https://www.sec.gov/Archives/edgar/data/933972/000165495426003094/scwo_10k.htm. Going-concern Note 3; ATM sales agreements; Nov 2024 offering terms; Mar 2026 convertible notes; "single significant stockholder" selling; AS system "six (6) and ten (10) tons per day"; AS1 "approximately 1 wet ton of waste per day."

[^W-4^]: 374Water Investor Presentation 4Q25 (company deck) — https://374water.com/wp-content/uploads/2026/03/374Water-Investor-Presentation-4Q25.pdf. System capacities (AS1 1 t/d @ 0.25–0.5 MGD; AS6 6 t/d @ 0.5–1.5 MGD; AS30 30 t/d @ 4–6 MGD; AS100+ @ 20–40 MGD); WDS site capex $1.5–2.0M/$4.5–6.0M/$7.5–10M; WDS revenue $2M+/$3–5M/$12–20M; "Identified demand totaling nearly $1.8BN"; ATM "~$7.0 million… runway into Q2 2026."

[^W-5^]: 374Water Q2 2026 results press release (Aug 18, 2026) — https://www.morningstar.com/news/accesswire/1208343msn/. "$2.26 million… approximately $2.0 million [from] the OC San Factory Acceptance Test… begin invoicing the remaining $2.6 million… Cash and cash equivalents: $1.8 million as of June 30, 2026… reduce its operating costs by approximately $3.2 million on an annualized basis."

[^W-6^]: 374Water Q2 2026 Form 10-Q, SEC EDGAR (accession 0001654954-26-007650, filed 2026-08-14) — https://www.sec.gov/Archives/edgar/data/933972/000165495426007650/scwo_10q.htm. Going concern ("cash balance of $1,776,259… accumulated deficit of $57,204,456"); Olathe PO "$4,880,000 firm fixed price, allocated $3,000,000 to the AirSCWO 6 unit…"; Orlando demo "$812,000"; OC San "costs in excess of billings of approximately $1.9 million"; subsequent events ($60k convertible note July 2026; ex-CFO demand letter).

[^W-7^]: 374Water 424B5 ATM prospectus, SEC EDGAR (filed 2026-08-11) — https://www.sec.gov/Archives/edgar/data/933972/000165495426007452/scwo_424b5.htm. "Up to $50,000,000 Common Stock… Robert's & Ryan… public float… approximately $27,000,000… approximately $1,142,000 shares of common stock remain available… under our Form S-3."

[^W-8^]: 374Water Q1 2026 results & highlights (8-K Ex-99.1, May 14, 2026) — https://www.sec.gov/Archives/edgar/data/933972/000165495426004906/scwo_ex991.htm. "purchase order and associated milestone payment of $2.2M from Garney… total contract is valued at more than $4.5M"; Orlando demo "approximately $482,000 in service revenue"; Q1 revenue $551,155, net loss $(4,571,623); USAF subcontract award.

[^W-9^]: DEF 14A special meeting proxy (reverse split), SEC EDGAR (Oct–Dec 2025) — https://www.sec.gov/Archives/edgar/data/933972/000165495425012524/scwo_def14a.htm. "to regain compliance with the Minimum Bid Price Requirement for continued listing on Nasdaq."

[^W-12^]: 374Water OC San deployment-timeline PR (Feb 13, 2025) — https://finance.yahoo.com/news/374water-announces-timeline-deployment-airscwo-212000017.html. "$1.3 Million in projected revenue…"; AS30 "30 wet tons of biosolids per day"; "up to five AirSCWO 200 systems." Insider sales table: "YAACOV NAGAR… 26 sales selling 1,322,292 shares for an estimated $1,294,373" (via Nasdaq/Quiver).

[^W-13^]: PowerVerde/374Water merger 8-K (Apr 2021) — https://www.sec.gov/Archives/edgar/data/933972/000157570521000209/pwvi_8k.htm. Founders' bios; Duke license; Merrell Bros. (MBH) $1,135,000 investment and Kokomo fabrication. Nagar settlement PR (Apr 10, 2025): https://finance.yahoo.com/news/374water-announces-settlement-claim-200100263.html.

[^W-14^]: 374Water company history — https://374water.com/company-history/; Wikipedia SCWO entry. Founded 2018 by Nagar & Deshusses as Duke spin-off; Gates Foundation “Reinvent the Toilet”; Nasdaq 2022; “Global Access” program.

[^W-16^]: Morningstar SCWO quote (Oct 2, 2026) — https://www.morningstar.com/stocks/xnas/scwo/quote. Close $2.31; 17.72M shares; market cap $41.81M.

[^W-17^]: TickerNerd SCWO statistics (Sept 24, 2026) — https://tickernerd.com/stock/scwo-forecast/. Market cap $54.4M; TTM revenue $1.9M; TTM net income −$20.0M; FCF −$11.3M; 48 employees; 17.7M shares.

[^W-18^]: 374Water DEF 14A (Aug 25, 2026), beneficial ownership table, SEC EDGAR — https://www.sec.gov/Archives/edgar/data/933972/000165495426007870/scwo_def14a.htm. “Yaacov Nagar 3,211,263 18.06%; Marc Deshusses 2,265,245 12.74%.”

[^W-20^]: Maven’s Notebook / Daily Pilot (Feb 2023) — https://mavensnotebook.com/2023/02/06/daily-digest-2-6-… “the district’s board of directors approved the project in 2021 at a cost of $8.7 million.”

[^W-21^]: 374Water Orlando 5-year WDS agreement PR (Mar 10, 2026) — https://www.nasdaq.com/press-release/374water-announces-5-year-agreement-city-orlando-approval-launch-licensed-waste. “the Company did not deliver the [OC San] system in 2025 as previously anticipated”; Orlando license 5-year + two 5-year extensions; >99.95% PFAS destruction.

[^W-22^]: OC San FAT completion PR (Jun 2, 2026) — https://finance.yahoo.com/sectors/energy/articles/374water-advances-solution-worlds-growing-120000117.html. “planned for deployment of the system and team in October of this year.”

[^W-23^]: B2i Digital 374Water profile (Aug 25, 2026) — https://b2idigital.com/374water. Orlando hub 88,000 gal storage, $3–5M/yr at initial scale; Orlando 40 MGD “would require more than thirty times the AirSCWO capacity now installed”; OC San first system “less than one percent”; Olathe 11 MGD “would require ten additional systems.”

[^W-24^]: St. Cloud campaign completion PR (Sep 15, 2026) — https://finance.yahoo.com/technology/articles/374water-completes-600-000-waste-120000104.html; Phase 2 PR (Jul 28, 2026). “$600,000 Waste Destruction Services (‘WDS’) contract with the City of St. Cloud, in partnership with Barr Engineering… supported by… LCCMR and… ENRTF”; “$500,000 and $1.5 million in annual revenue per unit.”

[^W-26^]: 374Water Q2 2025 earnings call coverage (Aug 13, 2025) — https://finance.yahoo.com/news/374water-inc-scwo-q2-2025-071207092.html. NC AFFF contract “valued at about $1 million, with potential for significantly higher value.”

[^W-27^]: DoD ESTCP/DIU Detroit demo completion PR (Oct 21, 2025) — https://finance.yahoo.com/news/374water-completes-commercial-scale-field-123100607.html. “approximately 900 gallons… over 80 hours of runtime” at Clean Earth’s Detroit TSDF; Arcadis prime.

[^W-28^]: Gaya Capital ENV Weekly (Aug 14, 2026) — https://www.gayacapital.com/envweekly/08142026. DIU “Success Memo formally validates 374Water’s AirSCWO… DoD estimates more than 700 military installations require PFAS remediation, with projected cleanup spending exceeding $3 billion.”

[^W-29^]: US Navy NDCEE selection PR (Dec 7, 2022) — https://finance.yahoo.com/news/374water-selected-us-navy-joint-140000148.html. Contract value undisclosed.

[^W-30^]: 374Water standard boilerplate (e.g., GlobeNewswire Feb 13, 2025) — “AirSCWO products destruction capacity scale from 1 ton per day to over 200 tons per day. 374Water offers capital sale and lease options, as well as turn-key destruction services.”

[^W-31^]: (don’t) Waste Water podcast profile of Kobe Nagar — https://dww.show/who/kobe-nagar/. “smallest containerized unit, sized for a community of about 6,000 people… at roughly 30,000-person scale the system becomes a net producer of electricity”; ~$23.8M raised to date.

[^W-32^]: 374Water strategy announcement (Sep 9, 2026) — https://markets.businessinsider.com/news/currencies/374water-unveils-strategy-to-scale-airscwo-into-a-global-waste-destruction-platform-1036533961. “regional infrastructure: centralized facilities capable of accepting biosolids from multiple wastewater treatment plants… 42,000 dry tons… at an estimated disposal cost of $500 to $1000… per dry ton”; Olathe “approximately $4.8 million PO.” ## General Atomics EMS (PERSES) — from research/comp_general_atomics.md

374Water customers & geography — from research/cust_374water.md

[^W-c1^]: 374Water PR, “5-Year Agreement with the City of Orlando” (Mar 10, 2026) — https://www.nasdaq.com/press-release/374water-announces-5-year-agreement-city-orlando-approval-launch-licensed-waste. “Signed a 5-year agreement with the City of Orlando, FL… includes two 5-year extensions.”

[^W-c2^]: 374Water PR, “Expands Orlando Partnership with Amended License Agreement” (Jun 9, 2026) — https://finance.yahoo.com/sectors/energy/articles/374water-expands-orlando-partnership-amended-120000402.html. “Add 88,000 gallons of onsite tank storage… total area to over 35,000 square feet… expected to generate between $3 million and $5 million in annual revenue.”

[^W-c3^]: 374Water strategy PR (Sep 9, 2026) — https://markets.businessinsider.com/news/currencies/374water-unveils-strategy-to-scale-airscwo-into-a-global-waste-destruction-platform-1036533961. “increasing capacity approximately 3-4 times, with a target of $10 million to $14 million in annual revenue”; “Projects… underway in Orlando, Florida; Orange County, California; St. Cloud, Minnesota; and Olathe, Kansas”; third horizon “expanding the WDS model internationally through a mix of company-owned facilities, licensing agreements, joint ventures and strategic partnerships.”

[^W-c6^]: 374Water Investor Presentation 4Q25 (Dec 2025) — https://374water.com/wp-content/uploads/2026/03/374Water-Investor-Presentation-4Q25.pdf. Deployment map: “AS6 Deployment Orange County Sanitation District Plant No.1 Fountain Valley, CA; AS Deployment Colorado School of Mines Peterson Space Force Base, CO; AS6 Deployment DIU/Arcadis/Clean Earth Hazardous Waste Center Detroit, MI; Waste Destruction Services RCRA TSDF OH; R&D&E… Morrisville, NC; Manufacturing and RD&E Facility Orlando, FL”; “Completed the Colorado School of Mines service project at Peterson Space Force Base.”

[^W-c9^]: 374Water Q2 2026 results PR (Aug 18, 2026) — https://www.morningstar.com/news/accesswire/1208343msn/. “OC San, California: Successfully completed the Factory Acceptance Test… with delivery to Orange County, California scheduled for October and unlocking an additional $2.6 million of contract value”; “Olathe, Kansas: Advancing our contracted $4.8M deployment.”

[^W-c11^]: Maven’s Notebook / Daily Pilot digest (Feb 2023) — cited in comp profile. “the district’s board of directors approved the project in 2021 at a cost of $8.7 million.”

[^W-c15^]: B2i Digital 374Water profile (Aug 25, 2026) — https://b2idigital.com/374water. “a $4,880,000 firm fixed price purchase order from Garney Companies for the Cedar Creek Wastewater Treatment Plant Expansion Phase II project… warranty coverage that runs 24 months from acceptance”; “delivery, commissioning and system operations are anticipated to begin in the last quarter of 2026 and continue through the first half of 2027” (OC San).

[^W-c16^]: 374Water Q2 2026 Form 10-Q, SEC EDGAR (filed Aug 14, 2026) — per comp profile [^6^]: Olathe PO “$4,880,000 firm fixed price, allocated $3,000,000 to the AirSCWO 6 unit.”

[^W-c17^]: 374Water Olathe award PR (Oct 28, 2025) — https://s3.amazonaws.com/b2icontent.irpass.cc/3021/rl158844.pdf. “the agreement, which is valued at approximately $4 million over the next two years.”

[^W-c18^]: 374Water Q1 2026 results PR (May 14, 2026) — https://www.morningstar.com/news/accesswire/1166970msn/. “building the next-generation AirSCWO system for contracted delivery to Olathe, KS in partnership with Garney Construction”; “awarded a subcontract… to support the U.S. Air Force in addressing PFAS contamination at firefighter training areas across military installations nationwide.”

[^W-c19^]: 374Water St. Cloud Phase 2/3 PR (Jul 28–29, 2026) — https://www.usatoday.com/press-release/story/38669/. “a $600,000 Waste Destruction Services (‘WDS’) partnership with Barr Engineering Co… at the City’s Nutrient, Energy, and Water Recovery Facility.”

[^W-c20^]: 374Water St. Cloud completion PR (Sep 15, 2026) — https://finance.yahoo.com/technology/articles/374water-completes-600-000-waste-120000104.html. “conducted under a $600,000 Waste Destruction Services (‘WDS’) contract with the City of St. Cloud, in partnership with Barr Engineering Co… supported by the Legislative-Citizen Commission on Minnesota Resources (‘LCCMR’) and the State’s Environment and Natural Resources Trust Fund (‘ENRTF’)”; “Each phase concluded with no operational issues”; mobile unit “potential to generate between $500,000 and $1.5 million in annual revenue per unit.”

[^W-c23^]: 374Water Detroit demo completion PR (Oct 21, 2025) — https://finance.yahoo.com/news/374water-completes-commercial-scale-field-123100607.html. “six weeks from July through September 2025… at a Clean Earth Treatment Storage and Disposal Facility (‘TSDF’) located in Detroit, MI… approximately 900 gallons of PFAS-laden feedstocks… over 80 hours of runtime… Arcadis, the project’s prime engineering contractor.”

[^W-c25^]: Proactive Investors (Apr 30, 2026) — https://www.proactiveinvestors.com/companies/news/1091523/. “Liquid waste streams achieved more than 99.9% destruction and removal efficiency (DRE), while solid waste streams exceeded 90%.”

[^W-c26^]: Prism Market View (Aug 25, 2026) — https://prismmarketview.com/374water-completes-phase-3-of-minnesota-pfas-destruction-pilot-as-army-corps-validation-builds-federal-momentum/. “a U.S. Army Corps of Engineers technical report dated July 2026, ERDC TR-26-19, documented that AirSCWO destroyed 99.9993% of PFAS in aqueous film-forming foam (AFFF)… across 42 compounds tested… Arcadis U.S…. co-authored the report.”

[^W-c27^]: Knoxnews/374Water PR, “374Water and Arcadis Team Receives Defense Dept. Success Memo” (Aug 13, 2026) — https://www.knoxnews.com/press-release/story/221479/. “The federal procurement pathway opened by the DIU Success Memo and the OTA accelerates 374Water’s ability to replicate this model at additional federal, commercial TSDF, or public-private partnership sites.”

[^W-c28^]: 374Water Peterson SFB PR (Jul 10, 2025) — https://www.nasdaq.com/articles/374water-deploys-airscwo-technology-dod-estcp-project-pfas-waste-treatment. “treat a concentrated PFAS liquid waste stream known as foam fractionate… generated at Peterson Space Force Base in Colorado Springs, Colorado… utilizes 374Water’s mobile AirSCWO system.”

[^W-c29^]: 374Water NC AFFF PR (Nov 4, 2025) — https://www.globenewswire.com/news-release/2025/11/04/3180355/0/en/374water-commences-contract-with-state-of-north-carolina-for-aqueous-film-forming-foam-destruction.html. “374Water will provide WDS to the University of North Carolina (‘UNC’) at Chapel Hill Collaboratory. North Carolina’s 60,000-gallon AFFF destruction initiative… first phase will treat 1,000 gallons… If selected for the second phase… up to an additional 28,000 gallons.”

[^W-c30^]: Yahoo Finance, 374Water Q2 2025 earnings coverage (Aug 13, 2025) — cited in comp profile. NC AFFF contract “valued at about $1 million, with potential for significantly higher value.”

[^W-c31^]: 374Water Q1 2025 earnings call transcript (May 15, 2025) — https://www.investing.com/news/transcripts/earnings-call-transcript-374water-q1-2025-sees-revenue-surge-stock-dips-93CH-4049364. “this multi phase contract could be worth north of $5,000,000 in future revenues” (NC AFFF); “Crystal Clean will not only host our [AirSCWO] system at the RCRA Part B TSDF in Oregon, Ohio, but also provide waste hauling, processing, storage, and other services.”

[^W-c32^]: 374Water/Crystal Clean hosting agreement PR (Sep 8, 2025) — https://www.globenewswire.com/news-release/2025/09/08/3146104/0/en/374water-announces-hosting-and-collaboration-agreement-with-crystal-clean.html. “374Water will establish a full-service WDS operation… at one of Crystal Clean’s facilities… the AirSCWO 6 system, which has the potential, over time, to generate $3 million to $5 million in annual recurring revenue. 374Water may install a second AirSCWO 6 system in 2026.”

[^W-c34^]: 374Water US Navy NDCEE PR (Dec 7, 2022) — https://finance.yahoo.com/news/374water-selected-us-navy-joint-140000148.html. Contract value undisclosed.

[^W-c36^]: Proactive Investors CEO interview (Oct 10, 2025) — https://www.proactiveinvestors.com/companies/news/1080140/. “In Australia, we have large opportunities, working with both the government and municipalities, as well as militaries. We see enormous opportunities in Europe… and also in Asia. We’re tracking many opportunities but remain focused on North America for now, with plans to expand internationally next year.”

[^W-c37^]: Proactive Investors (Sep 9, 2026) — https://www.proactiveinvestors.com/companies/news/1098296/. “The company’s third horizon involves expanding the WDS model internationally through a mix of company-owned facilities, licensing agreements, joint ventures and strategic partnerships, depending on the region.”

General Atomics EMS (PERSES) — from research/comp_general_atomics.md

[^G-1^]: **Forbes — “Neal Blue & family” profile** (accessed Sep. 2026; page dated 2026-09-15). https://www.forbes.com/profile/neal-blue/ — *“With his brother Linden, he bought the then nuclear energy company from Chevron for a reported $60 million in 1986… The largest private government contractor in the U.S.”*

[^G-3^]: **fed-spend.com — “General Atomics Revenue and Federal Contracts: $41.5B”** (2026-04-13). https://fed-spend.com/blog/general-atomics-federal-contracts-deep-dive — *“Privately held by the Blue family since 1986… It reported $3.2 billion in revenue in 2024 with approximately 15,000 employees… $41.5 billion in cumulative federal contract obligations, including a single $14.1 billion IDIQ awarded in September 2025… industry analysts estimate the company would be valued at $15-25 billion if it were publicly traded.”*…

[^G-4^]: **Forbes — General Atomics company overview** (accessed 2026). https://www.forbes.com/companies/general-atomics/ — *“General Atomics Financial Summary… Revenue $3.2B. Sources. FactSet; Forbes.”*

[^G-8^]: **Rheinmetall press release** (2026-06-18). https://www.rheinmetall.com/en/media/news-watch/news/2026/06/2026-06-18-rheinmetall-and-general-atomics-explore-co-production-of-vektrex-for-artillery-modernisation — *“GA-EMS is one of four main business segments of General Atomics… specializes in electromagnetic aircraft launch systems (EMALS) and advanced arresting gear (AAG), power generation and energy storage, space systems, nuclear fission energy systems… GA-EMS also provides commercial products and services tar…

[^G-9^]: **GA-EMS presentation (John Follin) to BAE/ARDEC, hosted by CSWAB (PDF)** (2016-12-13). https://cswab.org/wp-content/uploads/2017/03/Supercritical-Water-Oxidation-BAE-Presentation-Dec-2016.pdf — *“22 years and $250M of SCWO development; Built over 15 working SCWO systems; Over 35,000 hours of steady state operations”*; *“3-gpm iSCWO Shipping Configuration 25 feet long by 8 feet wide by 9.5 feet tall. Approximate weight = 20,800 pounds”*; *“Multiple systems can be added in parallel to increase capacity (x 3GPM)”*…

[^G-10^]: **NDIA Future Force Capabilities Conference — John Follin (GA-EMS), “PFAS Waste Destruction using GA’s iSCWO” (PDF)** (2025-10-02). https://www.ndia.org/-/media/sites/ndia/meetings-and-events/proceedings/2025/FFC/October%202/Follin_John_%202025%20FFC_2%20October%202025.pdf — *“1990s-2012 Development & Select US Government Uses; 2012-2014 First commercial sale…; Won five large Government PFAS test programs along with continued organic destruction sales. Introduced PERSES name for iSCWO; First two systems are stil…

[^G-11^]: **National Research Council — “Evaluation of Demonstration Test Results of Alternative Technologies for Demilitarization of Assembled Chemical Weapons” (Ch. 3, General Atomics Technology Package)** (2000). https://www.nationalacademies.org/read/9776/chapter/5 — *“Although the SCWO system successfully destroyed organic compounds in the liquids, the results did not demonstrate that the system is capable of operating without frequent shutdowns for repair or cleaning… corrosion and salt plugging both raised concerns…

[^G-13^]: **GA press release — “General Atomics to Demonstrate PFAS Destruction System at DIU-Selected Clean Earth Facility in North Carolina”** (2025-01-07). https://www.ga.com/ga-to-demonstrate-pfas-destruction-system-at-diu-selected-clean-earth-facility-in-north-carolina — *“The Defense Innovation Unit (DIU) has selected the Clean Earth waste facility in Charlotte, North Carolina… Under a previously awarded contract from the DoD Environmental Security Technology Certification Program (ESTCP) through the DIU… The Clean…

[^G-14^]: **GA-EMS standard boilerplate (e.g., Morningstar/ACCESS Newswire)** (2024-12-02). https://www.morningstar.com/news/accesswire/949217msn/general-atomics-awarded-navy-contract-to-advance-long-range-maneuvering-projectile — *“GA-EMS offers an expanding portfolio of capabilities for defense, government, and national security customers. GA-EMS also provides commercial products and services targeting the hazardous waste remediation, oil and gas, and nuclear energy industries.”*

[^G-15^]: **GA press release — “General Atomics Demonstrates PFAS Waste Destruction for Invited Federal, State and Waste Management Representatives”** (2023-09-12). https://www.ga.com/iscwo-pfas-destruction-demo-day-success — *“To date, our iSCWO systems have destroyed over 6 million gallons of hazardous and non-hazardous waste with a greater than 99.99% destruction efficiency.”*

[^G-16^]: **GA press release — “General Atomics Introduces PERSES Waste Destruction System”** (2025-03-10). https://www.ga.com/ga-introduces-perses-waste-destruction-system — *“PERSES™, the new name for the company’s commercial waste destruction system formerly known as the iSCWO… PERSES offers greater throughput capacity… Unlike many lab and prototype systems offered today, GA-EMS has had commercial systems in the field since 2012 and has successfully tested the system to destroy more than 200 waste streams.”* Two 2025 s…

[^G-17^]: **GA press release — “General Atomics’ iSCWO Process Tested, Verified by EPA For the Destruction of PFAS”** (2023-04-24). https://www.ga.com/ga-iscwo-process-tested-verified-by-epa-for-the-destruction-of-pfas — *“The Environmental Protection Agency (EPA) issued a detailed report (EPA/600/R-22/257) documenting test results and verifying GA-EMS’ commercial iSCWO system’s effectiveness in the destruction of PFAS… greater than 99.99%… We have been advancing SCWO technology since the 1990s, supplying systems to suppo…

[^G-19^]: **U.S. EPA — “Alternative Treatment Technologies to Open Burning and Open Detonation of Energetic Hazardous Wastes” (final report, Dec. 2019)**. https://www.epa.gov/sites/production/files/2019-12/documents/final_obod_alttechreport_for_publication_dec2019_508_v2.pdf — *“Alkaline hydrolysis… has been in production at Tooele Army Depot, Utah, for many years… The iSCWO neutralization technology was added to treat secondary waste streams resulting from the base hydrolysis process but eventually was not used due to la…

[^G-20^]: **Wikipedia — Blue Grass Chemical Agent-Destruction Pilot Plant** (accessed Sep. 2026). https://en.wikipedia.org/wiki/Blue_Grass_Chemical_Agent-Destruction_Pilot_Plant — *“In August 2020, the decision was made not to use the supercritical water oxidation system to process hydrolysate… In March 2021, program officials announced that hydrolysate… would be destroyed at Veolia North America near Port Arthur, Texas.”*

[^G-21^]: **National Research Council — “Analysis of Engineering Design Studies for Demilitarization of Assembled Chemical Weapons at Blue Grass Army Depot” (Ch. 5)** (2002). https://nap.nationalacademies.org/read/10509/chapter/7 — BGCAPP design with four SCWO reactors and brine-recovery system; off-site disposal of SCWO effluent under investigation.

[^G-22^]: **NATO/CCMS SCWO fact sheet (PDF)** (2002-12-11). https://www.ihpa.info/docs/library/reports/nato/NATO_SCWOFactSh&Annex10.pdf — *“A commercial scale 2 gpm General Atomics demonstration system is operating in Japan under a joint development agreement with… Kurita Water Industries, Ltd and Komatsu, Ltd… operated more than 1000 hours and at commercial scale at 1 ton/hour (8 t/day)… Pilot tests up to 227 kg/hr. Full-scale Blue Grass design is 2700 kg/hr.”*

[^G-23^]: **UNEP/Basel Convention — “General technical guidelines on the environmentally sound management of POPs wastes” (UNEP/CHW.15/6/Add.1; also COP17 revision)** (English PDF, accessed 2026). https://www.basel.int/Portals/4/download.aspx?d=UNEP-CHW.15-6-Add.1-Rev.1.English.pdf — *“General Atomics Electromagnetic Systems (GA-EMS) have developed an industrial SCWO system (iSCWO) which [is] transportable and housed in shipping containers… Transportable iSCWO is currently used in Kyrgyzstan and Tajikistan to destroy the…

[^G-24^]: **GA press release — “General Atomics Awarded Contract for Hazardous Waste Destruction Systems in China”** (2018-04-17). https://www.ga.com/general-atomics-awarded-contract-for-hazardous-waste-destruction-systems-in-china — *“Awarded a*

contract from Tianjin Saipeng Environmental Protection Technology Company… for three iSCWO systems… installed at the Binhai New Area facility."

[^G-25^]: **GA press release — "Defense Innovation Unit Awards General Atomics a Contract to Demonstrate iSCWO System for PFAS Destruction"** (2023-12-11). https://www.ga.com/ga-awarded-defense-innovation-unit-contract-to-demonstrate-iscwo-system-for-pfas-destruction — *"GA-EMS will deliver and install a commercial-scale iSCWO system at a DoD-identified site to conduct on-site PFAS destruction testing starting in early 2024."*

[^G-27^]: **GA press release — "General Atomics PERSES Waste Destruction System Scheduled to Destroy PFAS at Lake Elmo, Minnesota Site"** (2025-04-16). https://www.ga.com/ga-perses-waste-destruction-system-scheduled-to-destroy-pfas-at-lake-elmo-minnesota-site — *"GA-EMS was awarded a contract by Bay West LLC… leading the research program funded by the U.S. Army Engineer Research and Development Center (ERDC)… The system will be at the Lake Elmo site for about 4 weeks and then removed to other sites for additional tests un…

[^G-28^]: **Waste Advantage Magazine — "General Atomics Demonstrates PFAS Destruction System for Minnesota Congresswoman, State and Local Officials"** (2025-06-10). https://wasteadvantagemag.com/general-atomics-demonstrates-pfas-destruction-system-for-minnesota-congresswoman-state-and-local-officials/ — confirms Bay West/ERDC program and on-site Lake Elmo demos.

[^G-31^]: **EnviroWiki — "Supercritical Water Oxidation (SCWO)"** (page updated 2026-03-02). https://www.enviro.wiki/index.php?title=Supercritical_Water_Oxidation_(SCWO) — *"Early projections for treatment costs (CAPEX + OPEX) for slurries are in the range of $12 to $90 per ton (or $0.04 to $0.37 per gallon) depending on system scale and contaminant concentration, with a majority of the cost coming from amortizing the equipment."* (Generic SCWO economics, not GA-specific pricing.)

[^G-32^]: **IPQwery — General Atomics trademark portfolio** (accessed 2026-01). https://www.ipqwery.com/ipowner/en/owner/ip/510712-general-atomics.html — PERSES USPTO serial 99221721, filed 2025-06-06, **registered 2026-01-13**, classes 7/11/37/40 (waste destruction machines, wastewater treatment apparatus, installation/repair, treatment of materials).

[^G-33^]: **Wikipedia — Supercritical water oxidation (commercial applications)** (accessed Sep. 2026). https://en.wikipedia.org/wiki/Supercritical_water_oxidation — *"Aquarden is also providing remediation of hazardous energetic wastes and chemical warfare agents with SCWO, where a full-scale SCWO system has been operating for some years in France for the Defense Industry."* Also Chematur/Johnson Matthey UK AquaCat unit (3,000 L/h, "the largest SCWO unit in the world"). No GA–ArianeGroup connection documented. ## Aquarden Technologies (SuperOx) — from research/comp_aquarden.md

General Atomics customers & geography — from research/cust_ga.md

[^G-c1^]: **U.S. EPA — "Alternative Treatment Technologies to Open Burning and Open Detonation of Energetic Hazardous Wastes"** (Dec. 2019). https://www.epa.gov/sites/default/files/2019-12/documents/final_obod_alttechreport_for_publication_dec2019_508_v2.pdf — *"Alkaline hydrolysis… has been in production at Tooele Army Depot, Utah… The iSCWO neutralization technology was added to treat secondary waste streams resulting from the base hydrolysis process but eventually was not used due to lack of capacity and reliability issues, and offsite treatment methods were found to be more cost effective."* Table: *"Tooele Army Depot (UT) destruction of CADs and PADs ceased operation in 2016 due to dwindling waste stream."*

[^G-c2^]: **Parsons — "U.S. Chemical Weapons Destruction – Richmond, Kentucky" project page** (page dated 2023-12-27). https://www.parsons.com/project/blue-grass-chemical-weapons-stockpile-destruction-richmond-kentucky/ — *"Project Value: $1.1B Parsons/$5.3B Joint Venture… The Blue Grass Chemical Agent Pilot Plant is a 73-acre site located within the Blue Grass Army Depot in Richmond, Kentucky."*

[^G-c5^]: **Headquarters, Department of the Army — Report to Congress, "Feasibility to Reuse the BGCAPP and BGAD Infrastructure"** (Aug. 2023). https://media-cdn.dvidshub.net/pubs/pdf_67488.pdf — *"As of May 31, 2023, cumulative construction costs total $2B, including more than $100 million invested in infrastructure and site improvements."* Also: *"Construction activities progressed… Utility Building, and SCWO Building."*

[^G-c6^]: **Wikipedia — Blue Grass Chemical Agent-Destruction Pilot Plant** (accessed 2026). https://en.wikipedia.org/wiki/Blue_Grass_Chemical_Agent-Destruction_Pilot_Plant — *"In August 2020, the decision was made not to use the supercritical water oxidation system to process hydrolysate… hydrolysate… would be destroyed at Veolia North America near Port Arthur, Texas."*

[^G-c8^]: **GA-EMS presentation (John Follin), BAE/ARDEC, hosted by CSWAB** (2016-12-13). https://cswab.org/wp-content/uploads/2017/03/Supercritical-Water-Oxidation-BAE-Presentation-Dec-2016.pdf — *"GA History at Camp Minden: Shipped a 3gpm iSCWO system with compressor to Camp Minden for testing… Began Assembling 10gpm unit at Camp Minden; All*

work stopped because of Explo System Bankruptcy; US Army requested 10gpm unit to be shipped to MCAAP… 3gpm unit still at Camp Minden."

[^G-c9^]: **KSLA — "6 Explo officials indicted related to 2012 explosion at Camp Minden"** (2016-09-01). https://www.ksla.com/story/32942090/ — *"the U.S. Army awarded Explo a contract on March 24, 2010 to dispose of 450,000 155 mm artillery propelling charges designated as M119A2 for $2,902,500… amended… March 6, 2012 to dispose of 1,350,000 charges for $8,617,500."*

[^G-c12^]: **GA press release — "Defense Innovation Unit Awards General Atomics a Contract to Demonstrate iSCWO System for PFAS Destruction"** (2023-12-11). https://www.ga.com/ga-awarded-defense-innovation-unit-contract-to-demonstrate-iscwo-system-for-pfas-destruction — *"awarded a contract funded by… ESTCP through the Defense Innovation Unit (DIU)… deliver and install a commercial-scale iSCWO system at a DoD-identified site."* (DIU OTA — no value disclosed; not in USASpending contract records.)

[^G-c14^]: **NDIA Future Force Capabilities Conference — John Follin (GA-EMS), "PFAS Waste Destruction using GA's iSCWO"** (2025-10-02). https://www.ndia.org/-/media/sites/ndia/meetings-and-events/proceedings/2025/FFC/October%202/Follin_John_%202025%20FFC_2%20October%202025.pdf — *"Testing at Clean Earth in Charlotte, NC — iSCWO destruction demonstration tests – Six PFAS waste streams (Aug–Sep 2025)… Liquid concentrate waste streams from Cyclopure, Allonia, and ECT2; AFFF Waste Stream; Resin beads and GAC (Willow Grove)"*; *"US Army Corps of Engineers Test Site (Lake Elmo, MN)… Bay West is the prime… Three liquid PFAS waste streams (May–June 2025)"*; *"Install onsite testing at Minnesota (Leachates and biosolids) • Install onsite testing in Georgia (liquid waste streams)"*; *"2012–2014 First commercial sale… First two systems are still in use"*; *"Meets US, European and Asian Pressure and Electrical code requirements; EAR 99 Exportable."*

[^G-c18^]: **GA press release — "General Atomics PERSES Waste Destruction System Scheduled to Destroy PFAS at Lake Elmo, Minnesota Site"** (2025-04-16). https://www.ga.com/ga-perses-waste-destruction-system-scheduled-to-destroy-pfas-at-lake-elmo-minnesota-site — *"GA-EMS was awarded a contract by Bay West LLC… leading the research program funded by the U.S. Army Engineer Research and Development Center (ERDC)… The system will be at the Lake Elmo site for about 4 weeks and then removed to other sites for additional tests under different contracts."*

[^G-c19^]: **Bay West — "Bay West Leads Cutting-Edge Field Demonstration to Destroy PFAS in Lake Elmo, Minnesota"** (page dated 2026-02-24). https://baywest.com/experience/bay-west-leads-cutting-edge-field-demonstration-to-destroy-pfas/ — *"Site preparation underway in Lake Elmo, Minnesota, where Bay West is leading a full-scale PFAS destruction field demonstration using supercritical water oxidation… Bay West will be overseeing full-scale field deployment of the General Atomics PERSES system."*

[^G-c20^]: **USASpending.gov — Award W912HZ23C0037** (API lookup, Oct. 2026). https://www.usaspending.gov/award/CONT_AWD_W912HZ23C0037_9700_-NONE-_-NONE- — Recipient BAY WEST LLC; description "PFAS DESTRUCTION RESEACH U433D10"; **total obligation $9,999,105**; period 2023-07-31 → 2026-01-31; awarding agency DoD (W912HZ = USACE ERDC, Vicksburg).

[^G-c23^]: **GA press release — "General Atomics Awarded Contract for Hazardous Waste Destruction Systems in China"** (2018-04-17). https://www.ga.com/general-atomics-awarded-contract-for-hazardous-waste-destruction-systems-in-china — *"awarded a contract from Tianjin Saipeng Environmental Protection Technology Company, People's Republic of China, for three industrial Supercritical Water Oxidation (iSCWO) systems. The iSCWO systems will be installed at the Binhai New Area facility in the Tianjin municipality… GA-EMS will build, test and ship three iSCWO systems."*

[^G-c25^]: **UNEP/GEF project page — "Demonstration of Non-thermal Treatment of DDT Wastes in Central Asia" (GEF ID 9421)** (start 2020-09-16; accessed 2026). https://www.unep.org/gef/projects/demonstration-non-thermal-treatment-ddt-wastes-central-asia — *"Co-financing: $29,062,033; Project Grant: $15,120,000; Total Value: $44,182,033… Country: Kyrgyzstan, Tajikistan… Start Date: 16 September 2020; End Date: 31 December 2027."*

[^G-c28^]: **UN Tajikistan — Request for Expressions of Interest, GEF 9421 project** (posted/closing 2025-12-09). https://tajikistan.un.org/en/306175-request-expression-interest-project-gef-9421-demonstration-non-thermal-treatment-ddt-waste — *"prepare design, technical specification and cost estimate documentation for the construction of approximately 2.4 km gravel/unsealed road and fencing… approximately 3.2 km in length for the Vakhsh chemical waste site… part of the infrastructure development required for the safe management of DDT waste."*

[^G-c30^]: **Basel Convention OEWG-14 — General technical guidelines on POPs wastes (draft updated 2024-06-27; note 2024-07-15)** (same text in UNEP/CHW.15/6/Add.1/Rev.1). https://www.basel.int/Portals/4/download.aspx?d=UNEP-CHW-OEWG.14-INF-5-Rev.1.English.pdf — "*General Atomics Electromagnetic Systems (GA-EMS) have developed an industrial SCWO system (iSCWO) which [is] transportable and housed in shipping containers for on-site treatment of hazardous wastes containing POPs. Transportable iSCWO is currently used in Kyrgyzstan and Tajikistan to destroy the DDT content in 5000 tonnes of wastes containing DDT."*

[^G-c33^]: **NATO/CCMS SCWO fact sheet** (2002-12-11). https://www.ihpa.info/docs/library/reports/nato/NATO_SCWOFactSh&Annex10.pdf — *"A commercial scale 2 gpm General*

Atomics demonstration system is operating in Japan under a joint development agreement with… Kurita Water Industries, Ltd and Komatsu, Ltd… operated more than 1000 hours and at commercial scale at 1 ton/hour (8 t/day).” (Note: text internally also cites ~500 kg/h demo throughput.)

[^G-c34^]: **GA press release — “General Atomics Demonstrates PFAS Waste Destruction for Invited Federal, State and Waste Management Representatives”** (2023-09-12). https://www.ga.com/iscwo-pfas-destruction-demo-day-success — *“our iSCWO systems have destroyed over 6 million gallons of hazardous and non-hazardous waste with a greater than 99.99% destruction efficiency”* — demos at *“the company’s dedicated, full-scale system test facility located in San Diego.”*

[^G-c36^]: **AxNano website / EPA docket attachment EPA-HQ-OLEM-2020-0527-0136** (accessed 2026). https://downloads.regulations.gov/EPA-HQ-OLEM-2020-0527-0136/attachment_1.pdf — *“AxNano contracted GA to test and validate GA’s iSCWO system’s capability to destroy PFAS waste streams from a variety of sources.”*

[^G-c38^]: **GA Supplier Day 2022 — GA-EMS Programs Overview** (2022). https://www.ga.com/procurement/supplier-day-pdf/3_Programs_Overview_120122.pdf — *“GA-EMS provides services to customers worldwide. San Diego, CA - Headquarters. London, England. Abu Dhabi, UAE. New Delhi, India. Tokyo, Japan… GA-EMS Has Products in 68 Countries and Offices on 5 Continents and Now Facilities in Eggersriet, Switzerland [GA-Synopta].”*

Aquarden Technologies (SuperOx) — from research/comp_aquarden.md

[^A-1^]: Proff.dk — AQUARDEN TECHNOLOGIES ApS, Regnskab (annual reports 2021–2025, CVR/Virk data). URL: https://www.proff.dk/regnskab/aquarden-technologies-aps/skc3a6vinge/producenter/GQE5V1I016D (accessed Sep 2026). Verbatim (DKK ’000): “Årets resultat: -11.177 / -10.886 / -10.029 / -4.343 / -13.025”; “Egenkapital i alt: -21.505 … -30.947”; “Antal ansatte: 13 / 13 / 8 / 10 / 11”; “Likvide midler … 2.706”; “Anden gæld … 76.769”.

[^A-2^]: cvrapi.dk — AQUARDEN TECHNOLOGIES ApS. URL: https://cvrapi.dk/virksomhed/dk/aquarden-technologies-aps/29151911 (accessed Sep 2026). Verbatim: “CVR-nummer 29151911. Startdato 31/10 - 2005 … Ejere, Cai Holding … Direktører, Zhuoyan Cai”.

[^A-3^]: Proff.dk — Roller, AQUARDEN TECHNOLOGIES ApS. URL: https://www.proff.dk/roller/aquarden-technologies-aps/skc3a6vinge/producenter/GQE5V1I016D (accessed Sep 2026). “Direktør: Zhuoyan Cai; Stifter: Zhuoyan Cai; Revisor: Dansk Revision Hillerød.”

[^A-5^]: Aquarden — History & timeline. URL: https://aquarden.com/about/timeline/ (2024). “2024: Second industrial scale SuperOx SCWO plant operational in France with ArianeGroup for PFAS destruction … 2018: First industrial scale SCWO plant installed in France with our partner ArianeGroup … 2005: Aquarden Technologies founded in Denmark.”

[^A-6^]: Aquarden — SuperOx® SCWO. URL: https://aquarden.com/superox-scwo/ (2024). “Throughput 5 tonnes/day per unit … Operates solely on electricity. No fossil co-fuels required … Modular design.”

[^A-7^]: IFAT exhibitor presentation — “SuperOx® – SCWO Aquarden Technologies”. URL: https://exhibitors.ifat.de/en/exhibitors-products/products-services/products-services-details/productsDetail/2080/ (accessed Sep 2026). Verbatim: “Family-owned, Est. 2005 … Denmark, Singapore … Commercialization via select partners in France and North America – partnership with ArianeGroup”; “Field deployable: onsite or containerized … One reactor: Up to 12 m3/day liquid wastes/slurries … Energy consumption: 80-300 kWh per m3 wastewater”…

[^A-8^]: Aquarden — Aerospace & Defence. URL: https://aquarden.com/applications/aerospace-defence/ (2024). “Aquarden has provided our SuperOx® SCWO system for two of ArianeGroup’s French sites: the first industrial scale SuperOx® unit was operational in 2018 … followed by a second unit in 2024 for PFAS destruction … Our partnership began in 2017 through ArianeGroup’s innovative remediation project ELIXIR™.”

[^A-9^]: ArianeGroup — “ArianeGroup revolutionizes industrial waste management with Elixir™” (9 Feb 2024). URL: https://www.ariane.group/en/news/arianegroup-elixir-revolutionizing-industrial-waste-management/. Verbatim: “in partnership with Danish start-up Aquarden Technologies”; “A facility such as this can treat between 100 and 150 tons of solid waste and around 1,500 m3 of liquid waste”; “contained within a 200 m2 area, and can be dismantled and moved in less than two weeks”; “ArianeGroup’s next Elixir™ facility will…

[^A-10^]: French Ministry for Ecological Transition — Plan eau, 3 ans après (Innov’Eau). URL: https://portail.documentation.developpement-durable.gouv.fr/pub/MPDOUV00268987-plan-eau-ans-apres-100-des-mesures-initiees-dont-m.html (accessed Sep 2026). Verbatim: “Développé par ArianeGroup, ELIXIR est un procédé d’oxydation sous eau supercritique (SCWO) qui détruit > 99,99% des PFAS … Evalué à un coût total de 5,03 M€, ce projet lauréat Innov’Eau a bénéficié d’une aide de 1,59 M€.”

[^A-11^]: Krause et al. (2022), “Supercritical Water Oxidation as an Innovative Technology for PFAS Destruction,” ASCE J. Hazardous, Toxic, and Radioactive Waste / PMC10428202 (US EPA ORD-directed study). URL:

https://pmc.ncbi.nlm.nih.gov/articles/PMC10428202/. Verbatim: “PFOS was reduced from 26.2 mg/L to 240 µg/L … from the Aquarden … demonstration”; “For the demonstration, Aquarden used a feed flow of 1 L/h and a reactor residence time of 60 s.”

[^A-12^]: State of Green — “SCWO technology eliminates 99.9% of PFAS from wastewater” (9 Mar 2020). URL: https://stateofgreen.com/en/news/scwo-technology-eliminates-999-of-pfas-from-wastewater/. “Arlanda in Sweden has identified PFAS as a pollutant in their soil and water, and has therefore engaged Aquarden Technologies to remove these compounds.”

[^A-13^]: State of Green — “PFAS Destruction Through Supercritical Water Oxidation” (11 Nov 2020). URL: https://stateofgreen.com/en/news/pfas-destruction-through-supercritical-water-oxidation/. “A project with Stockholm Arlanda Airport similarly exhibited SCWO’s destruction efficiency with PFAS from a concentrated feed containing 679 µg/L PFOS and 34 µg/L PFOA, where both substances were eliminated to below their detection limits of 0.01 µg/L.”

[^A-15^]: Danish Environmental Protection Agency (MUDP) — “PFAS-inator: Sustainable and cost-effective technology for PFAS-removal from landfill leachate,” final report (Dec 2024; revised Jun 2026, ISBN 978-87-7038-692-0). URL: https://groenprojektbank.dk/Media/639183181296586761/Slutrapport%20-%20PFAS-inator.pdf. Verbatim: “under the Danish Environmental Agency’s grant ‘MUDP 2020’ … executed between February 2020 and December 2023”; “Aquarden’s fullscale SuperOx® plant can typically handle continuous treatment flowrates…

[^A-16^]: VIO.nu — PFAS-håndbogen (Danish PFAS handbook, 2022). URL: https://www.vio.nu/wp-content/uploads/2023/03/PFAS-haandbogen_2022-1.pdf. Verbatim (transl.): SCWO “is used inter alia in the Korsør case … also being trialled for leachate treatment at Danish landfills. The method is expensive and is therefore used as the final step in a treatment train…”

[^A-17^]: CleantechWatch — “Underskudsramt cleantech-selskab vil skabe forretning på PFAS-forurenet spildevand” (27 Sep 2021). URL: https://ctwatch.dk/nyheder/milj_teknik/article13305844.ece.

[^A-18^]: KFF Health News — “Inside the Pentagon’s Painfully Slow Effort to Clean Up Decades of PFAS Contamination” (21 Dec 2023). URL: https://kffhealthnews.org/public-health/pentagon-defense-department-pfas-contamination-cleanup-slow/. Verbatim: “said Zhuoyan Cai, director of Denmark-based Aquarden Technologies, which he said is currently in talks with U.S. companies about its SCWO technology.”

[^A-19^]: nano tech 2026 seminar — PFAS Busters LLC / Dr. Hiroshi Tomimasu (29 Jan 2026). URL: https://unifiedsearch.jcdbizmatch.jp/nanotech2026/en/sem/nanotech_mems/seminar_details/jnjY4V17wjU. “founded PFAS Busters LLC to introduce Aquarden Technologies’ Supercritical Water Oxidation (SCWO) technology from Denmark to Japan”; “The SuperOx system has been deployed in more than five countries in Europe.”

[^A-20^]: REGENESIS / Environment Analyst — “Corporate Guide: Identifying and Remediating PFAS” (2025). URL: https://regenesis.com/wp-content/uploads/2025/08/Environment-Analyst-Corporate-Guide-Identifying-and-Remediating-PFAS-2025.pdf. “SCWO units, such as those developed by General Atomics, Aquarden and 374Water, are at commercial launch stages. But they require high capital investment and skilled operation, often limiting their early adoption to larger regional facilities or remediation contractors.”

[^A-21^]: Sci-Bot — Global PFAS industry market assessment (26 Jun 2026). URL: https://sci-bot.ru/conduct-comprehensive-market-assessment-of-578a. “high capital cost (~USD 2–10M per unit)”; “USD 100–500/ton of concentrate”; “Aquarden demonstrated >99% PFAS removal for Swedish landfill leachate and Stockholm Arlanda Airport AFFF concentrate.”

[^A-27^]: ArianeGroup — ELIXIR™ datasheet (Apr 2026). URL: https://www.ariane.group/app/uploads/2026/06/datasheet-elixir-en-version2026_prov-basseresolution.pdf. “END-TO-END SERVICE … FIXED OR MOBILE FACILITIES, DIRECT PURCHASE OR LEASING.”

[^A-29^]: Aquarden — brochure download page (2026). URL: https://aquarden.com/download/. “The Aquarden SuperOx® enables fire services, airports, military sites, and industrial facilities to permanently remove PFAS e.g. from legacy firefighting foam stocks…” ## Revive Environmental (PFAS Annihilator) — from research/comp_revive.md

Aquarden customers & geography — from research/cust_aquarden.md

[^A-c1^]: Aquarden — Aerospace & Defence. URL: https://aquarden.com/applications/aerospace-defence/ (2024). Verbatim: “Aquarden has provided our SuperOx® SCWO system for two of ArianeGroup’s French sites: the first industrial scale SuperOx® unit was operational in 2018 for destruction of hazardous energetic wastes, followed by a second unit in 2024 for PFAS destruction … Our partnership began in 2017 through ArianeGroup’s innovative remediation project ELIXIR™.”

[^A-c2^]: CAIA — “L’innovation ArianeGroup” (Le magazine des ingénieurs de l’armement, June 2022; accessed 2026). URL: https://www.caia.net/revue-auteurs-rubriques-numeros/article/l-innovation-arianegroup/1036. Verbatim: “Une unité de traitement de produits pyrotechniques est actuellement opérée industriellement sur le site ArianeGroup de Saint-Médard-en-Jalles … a d’ores et déjà permis de détruire des déchets pyrotechniques tels que du propergol homogène (ROXEL), du premix (EURENCO), de la poudre B (SNPE Remédiation), de la nitrocellulose, des explosifs (HMX, …)”; “La capacité de destruction est de 100 à 200 tonnes de produits pyrotechniques détruits par an pour un réacteur.”

[^A-c3^]: Aquarden Technologies ApS annual report FY2023 (Danish CVR filing, via regnskaber.cvrapi.dk). URL: https://regnskaber.cvrapi.dk/11915192/amNsb3VkczovLzAzLzUxL2IxLzAyLzhhL2FlNTctNGRhZC1hMzc2LTJkMzI1ZjExNzQ1Zg.pdf. Verbatim: “Aquarden has in June 2024 just delivered a full-scale ‘SuperOx’ SCWO system to ArianeGroup in Toulouse as Europe’s first SCWO destruction facility for concentrated PFAS wastes … contracts have been secured for treating PFAS wastes at the Toulouse site”; “We have also been asked by large wastewater treatment plants in the USA for collaboration in hazardous sludge destruction.”

[^A-c4^]: Mediacités Toulouse — “L’usine Ariane va traiter des déchets de PFAS à Toulouse” (19 Jan 2026). URL: https://www.mediacites.fr/environnement/toulouse/2026/01/19/arianegroup-va-traiter-des-dechets-de-pfas-pres-de-lancien-site-dazf-a-toulouse/. Verbatim: “À l’été 2025, la préfecture de Haute-Garonne a autorisé … l’industriel ArianeGroup à installer une unité de traitement des PFAS sur [le site] … île d’Empalot … toujours classée Seveso seuil haut”; CSS disclosure 26 Sep 2025. (Partially paywalled; headline/lead extracted.)

[^A-c5^]: L’Usine Nouvelle — “ArianeGroup s’en sert pour ses systèmes de missiles M51 : la solution Elixir détruit les PFAS de l’eau grâce à l’eau supercritique” (25 Jul 2026). URL: https://www.usinenouvelle.com/recyclage-et-services-a-l-environnement/eau/arianegroup-sen-sert-pour-ses-systemes-de-missiles-m51-la-solution-elixir-detruit-les-pfas-de-leau-grace-a-leau-supercritique.APUO6JHM2RNTTDRML5VDNXDA3I.html. Verbatim: “Lancé en janvier 2025, le projet est doté d’un investissement de 5,03 millions d’euros, dont 1,59 million d’aides publiques.”

[^A-c6^]: French Ministry for Ecological Transition — “Plan eau : 3 ans après” (Innov’Eau focus, 2026). URL: https://portail.documentation.developpement-durable.gouv.fr/pub/MPDOUV00268987-plan-eau-ans-apres-100-des-mesures-initiees-dont-m.html. Verbatim: “Développé par ArianeGroup, ELIXIR est un procédé d’oxydation sous eau supercritique (SCWO) qui détruit > 99,99% des PFAS … Evalué à un coût total de 5,03 M€, ce projet lauréat Innov’Eau a bénéficié d’une aide de 1,59 M€.”

[^A-c10^]: ArianeGroup — “ELIXIR™: the ArianeGroup solution to decontaminate firefighting vehicles” (5 Dec 2025 / 9 Jan 2026). URL: https://www.ariane.group/en/news/elixir-a-solution-for-firefighting-vehicles-2/. Verbatim: “In June 2025, the SDIS from the Charente department (16) and the SDIS from the Charente-Maritime department (17) were selected by the 2025 Fonds Innov’Achats programme of the French Ministry of Interior…”; “Operations conducted on 23 firefighting vehicles … at the ArianeGroup Saint-Médard-en-Jalles (Issac) facilities”; “waste streams are sent to the ArianeGroup facilities in Toulouse in order to be destroyed by supercritical hydrothermal oxidation in the ELIXIR™ reactor” (Thibault Derode); partner “Curium.”

[^A-c11^]: State of Green — “SCWO technology eliminates 99,9% of PFAS from wastewater” (9 Mar 2020). URL: https://stateofgreen.com/en/news/scwo-technology-eliminates-999-of-pfas-from-wastewater/. Verbatim: “An initial study with leachate from Sörab waste deposit (SE)…”; “Arlanda in Sweden has identified PFAS as a pollutant in their soil and water, and has therefore engaged Aquarden Technologies to remove these compounds.”

[^A-c12^]: State of Green / WaterOnline — “PFAS Destruction Through Supercritical Water Oxidation” (10–11 Nov 2020). URL: https://stateofgreen.com/en/news/pfas-destruction-through-supercritical-water-oxidation/ and https://www.wateronline.com/doc/pfas-destruction-through-supercritical-water-oxidation-0001. Verbatim: “Aquarden received leachate for SCWO treatment and managed to reduce the PFAS levels from 3.7 µg/L to below 0.035 µg/L. A project with Stockholm Arlanda Airport … concentrated feed containing 679 µg/L PFOS and 34 µg/L PFOA … eliminated to below their detection limits of 0.01 µg/L.”

[^A-c13^]: Mellifiq — “Mellifiq Awarded Contract for Treating PFAS at Arlanda Airport” (24 Apr 2025). URL: https://mellifiq.com/en/mellifiq-awarded-contract-for-treating-pfas-at-arlanda-airport/. Verbatim: “Mellifiq is proud to announce that we have been awarded another significant project by Swedavia to deliver, install, and operate a water treatment system aimed at reducing PFAS concentrations at Arlanda Airport … 20,000 m³ per year, with a consistent flow rate of 6.5–7 m³/h.”

[^A-c15^]: Krause et al. (2022), “Supercritical Water Oxidation as an Innovative Technology for PFAS Destruction,” ASCE J. Env. Eng. / PMC10428202 (US EPA ORD study). URL: https://pmc.ncbi.nlm.nih.gov/articles/PMC10428202/. Verbatim: “Three different organizations were contracted … Battelle (Columbus, USA), Aquarden Technologies (Skaeving, Denmark), and 374Water (Durham, USA)”; “For the demonstration, Aquarden used a feed flow of 1 L/h and a reactor residence time of 60 s”; “ALS Environmental Lab analyzed a single effluent sample for 12 PFAS (Copenhagen, Denmark)”; “PFOS was reduced from 26.2 mg/L to 240 µg/L, 30.4 mg/L to 0.310 µg/L, and 190 mg/L to 8.57 µg/L, from the Aquarden, Battelle, and 374Water demonstrations, respectively.”

[^A-c16^]: Folketinget — MUDP beretning 2020, MOF Alm.del Bilag 476. URL: https://www.ft.dk/samling/20201/almdel/mof/bilag/476/2369480.pdf. Verbatim: “MUDP har støttet med 2.801.385 kr. ud af et samlet budget på 4.853.398 kr. Projekt: PFAS-inator: Bæredygtig og kosteffektiv teknologikoncept til fjernelse …”

[^A-c17^]: Danish EPA (MUDP) — PFAS-inator final report (Dec 2023/2024). URL: https://groenprojektbank.dk/Media/639183181296586761/Slutrapport%20-%20PFAS-inator.pdf. Verbatim: “Aquarden Technologies ApS, Vandrensning.com ApS, H2oNordic Denmark ApS, Eden Aquatech AB, Argo I/S (Audebo Landfill), Fors A/S, Perpetuum AS, Norge, Nomi4S I/S, Hjerring Vandselskab A/S, DTU Sustain and the Danish Technological Institute participated in

the project. The project was executed between February 2020 and December 2023”; followed by “Forsvarsministeriets Ejendomsstyrelse” and “Miljøstyrelsen.”

[^A-c19^]: Danish EPA (MST) — PFAS remediation-experiences publication (Apr 2025), ISBN 978-87-7038-737-8. URL: https://www2.mst.dk/Udgiv/publikationer/2025/04/978-87-7038-737-8.pdf. Verbatim: “I Danmark har virksomheden Aquarden Technologies udviklet et kommercielt SCWO-anlæg, som har været afprøvet med succes til destruktion af PFAS i vandigt affald fra oprydningssagen på Korsør Brandskole samt hos Ariane Group i Frankrig”; “Der findes ingen SCWO-anlæg i Danmark til modtagelse af PFAS-affald. Aquarden Technologies har solgt et anlæg til virksomheden Ariane Group i Frankrig … Anlægget har en kapacitet på 5 m³/døgn. I øjeblikket er Aquarden Technologies det eneste firma i Europa, der producerer og sælger kommercielle SCWO-anlæg.”

[^A-c20^]: IFAT exhibitor presentation — “SuperOx® – SCWO Aquarden Technologies”. URL: https://exhibitors.ifat.de/en/exhibitors-products/products-services/products-services-details/productsDetail/2080/ (accessed 2026). Verbatim: “Pharmaceuticals • Treatment of wastewater from Royal Copenhagen Hospital with more than 60 pharmaceuticals • 28 listed as hazardous by Danish EPA.”

[^A-c21^]: nano tech 2026 / InterAqua 2026 seminar listing — PFAS Busters LLC, CEO Dr. Hiroshi Tomimasu (29 Jan 2026, Tokyo). URL: https://unifiedsearch.jcdbizmatch.jp/nanotech2026/en/sem/nanotech_mems/seminar_details/jnjY4V17wjU. Verbatim: “founded PFAS Busters LLC to introduce Aquarden Technologies’ Supercritical Water Oxidation (SCWO) technology from Denmark to Japan, with the aim of addressing PFAS-related issues”; “The SuperOx system has been deployed in more than five countries in Europe.”

[^A-c23^]: KFF Health News (21 Dec 2023). URL: https://kffhealthnews.org/public-health/pentagon-defense-department-pfas-contamination-cleanup-slow/. Verbatim: “said Zhuoyan Cai, director of Denmark-based Aquarden Technologies, which he said is currently in talks with U.S. companies about its SCWO technology.”

[^A-c24^]: IFAT exhibitor deck (accessed 2026). Verbatim: “Family-owned, Est. 2005 … Denmark, Singapore … Commercialization via select partners in France and North America – partnership with ArianeGroup.”

Revive Environmental (PFAS Annihilator) — from research/comp_revive.md

[^R-1^]: Business Wire / Battelle — “Battelle, Viking Global Investors Launch Revive Environmental,” Jan 17, 2023. https://www.battelle.org/insights/newsroom/press-release-details/battelle-viking-global-investors-launch-revive-environmental (also https://revive-environmental.com/battelle-viking-global-investors-launch-revive-environmental-2023/)

[^R-2^]: Signal Net — “Quiet Breakout: The Strongest Bond in Organic Chemistry Has Six New Ways to Break,” Jun 5, 2026. https://www.signalnet.ai/quiet-breakout-the-strongest-bond-in-organic-chemistry-has-six-new-ways-to-break/

[^R-5^]: BlueTech Research Innovation Tracker — “Revive Environmental” (snapshot: Service Based/Treatment Fee; Private; Seed; Early Revenue). https://www.bluetechresearch.com/tools/innovation-tracker/revive-environmental/

[^R-6^]: GlobeNewswire — “Revive Environmental Announces CEO Transition,” Mar 6, 2026. https://finance.yahoo.com/news/revive-environmental-announces-ceo-transition-220800443.html

[^R-8^]: Revive Environmental blog citing Associated Press — “PFAS is Everywhere—And the World is Paying Attention” (AP interview, Mar 2025). https://revive-environmental.com/pfas-is-everywhere-and-the-world-is-paying-attention/

[^R-9^]: Democracy in NH / NH Bulletin — NH-Revive contract $668,258 for 10,000 gal AFFF; Heritage-Crystal Clean hauls to Wyoming, MI. Aug 2023. https://democracynh.com/?page=32 ; https://newhampshirebulletin.com/2023/07/25/nh-is-first-state-to-partner-with-pfas-annihilator-to-destroy-firefighting-foams/

[^R-10^]: Office of Ohio Governor — “Governor DeWine Announces Expansion of Firefighting Foam Takeback Program” (Round 1: $3M Monsanto PCB settlement; ≥60,000 gal remaining; registration to Sept 4, 2026). https://governor.ohio.gov/wps/portal/gov/governor/media/news-and-media/governor-dewine-announces-expansion-of-firefighting-foam-takeback-program

[^R-11^]: Gaya Capital ENV Weekly, May 1, 2026 — “New Jersey / Revive Environmental | $16.6m program appropriation; contract value not disclosed | ~150,000 gallons of AFFF from more than 400 departments.” https://www.gayacapital.com/envweekly/05022026

[^R-13^]: GlobeNewswire — “Revive Environmental First to Complete Verified PFAS Destruction Under North Carolina’s AFFF Take-Back Program,” Jan 22, 2026. https://www.globenewswire.com/news-release/2026/01/22/3224234/0/en/

[^R-14^]: GlobeNewswire — “EPA Administrator Lee Zeldin Tours Revive Environmental’s Columbus Headquarters, the Nation’s First Commercial PFAS Destruction Facility,” Jul 7, 2026 (5 Annihilator units; “cost-effective at scale”). https://revive-environmental.com/epa-administrator-lee-zeldin-tours-revive-environmentals-columbus-headquarters-the-nations-first-commercial-pfas-destruction-facility/

[^R-16^]: Business Wire — “Revive Environmental PFAS Annihilator Deployed in First-to-Market Commercial Destruction of ‘Forever Chemicals’,” May 1, 2023 (Grand Rapids; 300–500 gpd concentrate; 160k gpd raw leachate). https://www.businesswire.com/news/home/20230501005063/en/

[^R-17^]: Battelle — “West Michigan facility operating first commercial-scale PFAS-destroying technology,” May 5, 2023 (Wyoming, MI; 4never; >100k gpd leachate). https://www.battelle.org/insights/newsroom/news-details/2023/05/05/west-michigan-facility-operating-first-commercial-scale-pfas-destroying-technology

[^R-18^]: Revive Environmental — “Revive Environmental’s Columbus Facility Now Fully Permitted to Treat PFAS” (second permitted US site; first in Ohio), 2026. https://revive-environmental.com/revive-environmentals-columbus-facility-now-fully-permitted-to-treat-pfas/

[^R-19^]: NPR — “A new tool is helping to annihilate dangerous PFAS in firefighting foam,” Mar 16, 2026 (Ohio collected ~14,000 gal spring 2024; Ohio EPA seeking funds for more). https://www.npr.org/2026/03/16/nx-s1-5714017/

[^R-20^]: Business Wire — “Revive Environmental Wins 2025 WEFTEC Innovative Technology Award for PFAS Annihilator SCWO,” Sep 16, 2025. https://www.businesswire.com/news/home/20250916378962/en/

[^R-22^]: Revive Environmental — “Ohio AFFF Takeback Program Complete,” Aug 11, 2026 (13,500+ gal; 120 fire departments/airports; $3M Monsanto settlement). https://revive-environmental.com/ohio-collects-and-destroys-large-quantities-of-firefighting-foam-after-takeback-program-throughout-the-state/

[^R-23^]: Carolina Connection (UNC) — “A state pilot program to destroy PFAS has entered its final phase,” Apr 23, 2025 (4 vendors: AxNano, Clean Earth, Revive, 374Water; 374Water projects $100–$400/gal). https://carolinaconnection.org/2025/04/23/a-state-pilot-program-to-destroy-pfas-has-entered-its-final-phase

[^R-24^]: Gaya Capital ENV Weekly, May 1, 2026 — “Colorado / Aquagga | Contract not disclosed; $40/gal program payment | … using HALT.” https://www.gayacapital.com/envweekly/05022026

[^R-25^]: Business Wire — “Revive Environmental and Battelle Partner on DoD-Funded Project to Eliminate Toxic ‘Forever Chemicals’” (ESTCP/DIU demo at Clean Earth, Detroit), Apr 2, 2025. https://www.businesswire.com/news/home/20250402919534/en/

[^R-26^]: Mitchell Williams Law — analysis of DoW Feb 26, 2026 report: >2M gal AFFF + 1.5M gal rinsate; AFFF use prohibition extended to Oct 1, 2026. https://www.mitchellwilliamslaw.com/detailed-analysis-of-options-for-the-destruction/disposal-of-covered-materials-containing-per-and-polyfluoroalkyl-substances-us-department-of-war-report

[^R-27^]: Waste Dive — “DOD guidance recommends hazardous waste landfills, incineration to manage PFAS,” Mar 17, 2026 (SCWO listed among “emerging options”). https://www.wastedive.com/news/dod-guidance-hazardous-waste-landfills-incineration-manage-pfa/814937/

[^R-28^]: Nasdaq/Heritage-Crystal Clean — “Heritage-Crystal Clean, Allonnia, Revive Environmental, and EPOC Enviro Launch 4never™,” Apr 5, 2024 (hub-and-spoke; fixed cost per gallon). https://www.nasdaq.com/press-release/heritage-crystal-clean-allonnia-revive-environmental-and-epoc-enviro-launch-4nevertm

[^R-29^]: Revive Environmental — “Revive Environmental’s Impact at the 2025 ARFF Symposium” (airport AFFF/rinsate projects; Certificate of Annihilation, EPA Method 1633). https://revive-environmental.com/revive-environmentals-impact-at-the-2025-arff-symposium/

[^R-31^]: GlobeNewswire — “Revive Environmental Ready to Support DoW PFAS Destruction Requirements Under New Federal Guidance,” Mar 11, 2026 (“state agencies, DoW installations, Fortune 500 industrials, and municipal utilities”; “nation’s only commercially permitted PFAS destruction facilities”). https://www.globenewswire.com/news-release/2026/03/11/3253844/0/en/

[^R-34^]: Revive Environmental — “Landfill Leachate Treatment and PFAS Destruction” (on-site concentration; concentrate shipped to permitted facility; plug-and-play deployment “within weeks”). https://revive-environmental.com/landfill-leachate-pfas-services/

[^R-35^]: MLive — “Forever no more: New tech ‘annihilates’ toxic PFAS chemicals,” Aug 15, 2023 (shipping-container units; 15–40 ¢/gal; 6 units in production, up to 25; Crystal Clean 11 candidate facilities). https://www.mlive.com/public-interest/2023/07/forever-no-more-new-tech-annihilates-toxic-pfas-chemicals.html

[^R-36^]: ES&T Water (peer-reviewed), 2025 — spent-media economics; centralized reactivation contracts limited to large facilities (>36,000 kg/changeout); Scenario A off-site recovery cost-effective for “small- to medium-scale or fragmented systems.” https://pmc.ncbi.nlm.nih.gov/articles/PMC12260480/

[^R-37^]: Wisconsin Examiner — “System to ‘annihilate’ PFAS chemicals deployed in Michigan,” Jun 27, 2023 (~500 gpd concentrate; ~160–165k gpd raw leachate). https://wisconsinexaminer.com/2023/06/27/system-to-annihilate-pfas-chemicals-deployed-in-michigan/

[^R-38^]: Revive Environmental via Water Journal (Korea) — EPA interim disposal guidance PR, Apr 2024 (“capacity to destroy up to 7 million gallons of PFAS laden waste annually… price competitive with incineration”). https://www.waterjournal.co.kr/news/articleView.html?idxno=74941

[^R-39^]: Ohio OBM contract document (Battelle/Revive AFFF takeback scope) — collection events limited to <1,000 gal per waste owner; larger volumes require separate arrangements. https://api.obm.ohio.gov/files/download/44/eb81621a-9451-4df6-a3cf-652064421a2b/inline

[^R-40^]: Waste Dive — “DOD-funded PFAS demo projects show promise for remediation and destruction,” Dec 1, 2025 (on-site more efficient/cost-effective than off-site due to logistics/permitting; Revive treated AFFF in Detroit). https://www.wastedive.com/news/dod-pfas-destruction-disposal-demos-waste/805991/

[^R-43^]: Minnesota PCA — “Groundbreaking study shows unaffordable costs of PFAS cleanup from wastewater” (small facilities’ per-pound costs >6× large facilities). https://www.pca.state.mn.us/news-and-stories/groundbreaking-study-shows-unaffordable-costs-of-pfas-cleanup-from-wastewater

[^R-44^]: Route Fifty — “Small water utilities cannot achieve PFAS cleanup on their own,” Sep 30, 2024 (95% of systems <10,000 customers; ≤3 FTE staff). https://www.route-fifty.com/infrastructure/2024/09/small-water-utilities-cannot-achieve-pfas-cleanup-their-own/399605/ ## Expanded non-dilutive funding channels — from research/grants_ngo.md

Revive customers & geography — from research/cust_revive.md

[^R-c1^]: New Hampshire Bulletin — “NH is first state to partner with ‘PFAS Annihilator’ to destroy firefighting foams,” Jul 25, 2023. https://newhampshirebulletin.com/2023/07/25/nh-is-first-state-to-partner-with-pfas-annihilator-to-destroy-firefighting-foams/ — *“Funding for the state’s $668,258 contract with Revive will come from two DES pots of money: hazardous waste cleanup funds and emerging contaminants funds.”* (Corroborated: Democracy in NH, Aug 19, 2023, https://democracynh.com/?page=32; Seacoast Online, Jul 26, 2023.)

[^R-c2^]: Revive Environmental blog — “Celebrating Success: Revive Environmental’s AFFF Takeback Initiative in New Hampshire, Ohio, and Beyond,” Aug 14, 2024. https://revive-environmental.com/celebrating-success-revive-environmentals-afff-takeback-initiative-in-new-hampshire-ohio-and-beyond/ — *“Registered and collected over 12,000 gallons of AFFF from over 125 fire departments across the state through 10 county-wide events.”*

[^R-c3^]: NHDES — “AFFF Take Back Program” program page (accessed 2026). https://www.pfas.des.nh.gov/pfas-products/firefighting-foam/afff-take-back-program — *“The program removed and properly disposed of approximately 10,000 gallons of AFFF collected from 104 fire departments.”*

[^R-c4^]: Ohio EPA — “Ohio AFFF Takeback Program” page (updated Sep 2026). https://epa.ohio.gov/divisions-and-offices/communications-and-outreach/AFFF/afff-takeback-program — *“funded with $3 million in settlement funding Ohio received as part of the state’s enforcement case against Monsanto… awarded to Battelle and its subcontractor, Revive Environmental.”*

[^R-c5^]: GlobeNewswire / Revive Environmental — “Ohio Collects and Destroys Large Quantities of Firefighting Foam After Takeback Program Throughout the State,” Aug 11, 2026. https://www.globenewswire.com/news-release/2026/08/11/3342503/0/en/ohio-collects-and-destroys-large-quantities-of-firefighting-foam-after-takeback-program-throughout-the-state.html — *“collected from 118 fire departments and two municipal airports and have now been destroyed at Revive Environmental’s commercial PFAS destruction facility in Columbus.”*

[^R-c7^]: Office of Ohio Governor — “Governor DeWine Announces Expansion of Firefighting Foam Takeback Program,” 2026. https://governor.ohio.gov/wps/portal/gov/governor/media/news-and-media/governor-dewine-announces-expansion-of-firefighting-foam-takeback-program — *“Governor DeWine is directing more than $13 million from Ohio’s recent DuPont settlement to Ohio’s Aqueous Film-Forming Foam (AFFF) Takeback Program.”*

[^R-c9^]: Revive Environmental — “North Carolina Selects Revive Environmental for Statewide PFAS Destruction in AFFF Take-Back Program,” Mar 2025. https://revive-environmental.com/north-carolina-selects-revive-environmental-for-statewide-pfas-destruction-in-afff-take-back-program/ — *“The initial phase encompasses 1,000 gallons of AFFF, with provisions to expand to approximately 60,000 gallons.”*

[^R-c10^]: GlobeNewswire — “Revive Environmental First to Complete Verified PFAS Destruction Under North Carolina’s AFFF Take-Back Program,” Jan 22, 2026. https://www.globenewswire.com/news-release/2026/01/22/3224234/0/en/ — *“completed the destruction of 1,000 gallons of AFFF as part of Phase 1… becoming the first provider to complete treatment and independent verification.”*

[^R-c11^]: NJDEP News Release 26_00320 / NJ DCA — “Sherrill Administration Highlights Success of Historic Program…,” Mar 20, 2026. https://dep.nj.gov/newsrel/26_00320/ and https://www.nj.gov/dca/news/news/2026/20260320.shtml — *“funded through a $16.6 million state appropriation”; “The DEP has contracted with Revive Environmental to manage the logistics, consolidation, storage, and certified destruction of collected AFFF materials. Collected foam will be transported to a Revive-operated treatment facility in Ohio.”*

[^R-c12^]: Waste Advantage Magazine — “New Jersey and Revive Environmental Lead Statewide Program…,” Mar 23, 2026. https://wasteadvantagemag.com/new-jersey-and-revive-environmental-lead-statewide-program-to-collect-and-destroy-pfas-

containing-firefighting-foam/ — Republic Services SVP Julia Arambula: *“the largest-ever AFFF collection and destruction program.”*

[^R-c13^]: CBS News — “New Jersey works to destroy firefighting foam laced with cancer-linked PFAS,” Jul 11, 2026 (Burlington County dateline; Jacobstown VFC, 580 gal). https://www.cbsnews.com/news/new-jersey-destroy-firefighting-foam-cancer-linked-pfas/ — *“The state has so far collected more than 150,000 gallons from fire stations. All of it has been trucked to Revive Environmental.”*

[^R-c14^]: Colorado DPHE — “PFAS Takeback Program” page (accessed 2026). https://cdphe.colorado.gov/PFAS/Takeback — *“$40 per gallon for unspent firefighting foam containing PFAS”; “evaluated the proposals and awarded a contract to Aquagga.”*

[^R-c15^]: Aquagga blog — “Aquagga Selected for Colorado’s PFAS Takeback Program,” May 6, 2026. https://www.aquagga.com/blog/aquagga-selected-for-colorados-pfas-takeback-program — *“This three-year contract will run through fall 2028… approximately 40,000 gallons of firefighting foam stored throughout Colorado.”*

[^R-c17^]: Business Wire — “Revive Environmental PFAS Annihilator™ Deployed in First-to-Market Commercial Destruction of ‘Forever Chemicals’,” May 1, 2023. https://www.businesswire.com/news/home/20230501005063/en/ — *“Heritage-Crystal Clean will work with multiple landfills… it will process between 300 to 500 gallons of concentrated landfill leachate per day (representing 300,000 to 500,000 gallons of raw leachate).”*

[^R-c18^]: MLive — “Forever no more: New tech ‘annihilates’ toxic PFAS chemicals,” Aug 15, 2023. https://www.mlive.com/public-interest/2023/07/forever-no-more-new-tech-annihilates-toxic-pfas-chemicals.html — *“The facility is currently processing leachate from three Michigan landfills”; “the treatment cost is running between 15 and 40 cents per-gallon”; “one [Battelle DoD contract] to eventually deploy an Annihilator unit at Wurtsmith Air Force Base in Oscoda.”*

[^R-c19^]: Crystal Clean — “How Crystal Clean Is Tackling PFAS at Grand Rapids,” Nov 12, 2024. https://www.crystal-clean.com/pfas-forever-chemicals-concentration-destruction-technology/ — three SAFF®40 + first US SAFF® Hyper-Concentrator; up to 300,000 gpd; *“PFAS impacted waters arrive at Grand Rapids from a variety of locations including landfills as well as industrial wastewaters, fire-impacted waters and more.”*

[^R-c22^]: Business Wire — “Revive Environmental and Battelle Partner on DoD-Funded Project to Eliminate Toxic ‘Forever Chemicals’,” Apr 2, 2025. https://www.businesswire.com/news/home/20250402919534/en/ — *“The field demonstration will showcase Revive Environmental’s PFAS Annihilator® at a Clean Earth facility in Detroit, MI.”*

[^R-c23^]: Waste Dive — “DOD-funded PFAS demo projects show promise for remediation and destruction,” Dec 1, 2025. https://www.wastedive.com/news/dod-pfas-destruction-disposal-demos-waste/805991/ — *“Revive Environmental, which treated AFFF in Detroit… ‘it did the job we were expecting it to.’”*

[^R-c24^]: Revive Environmental — “Transitioning ARFF Vehicles Away from PFAS: Fort Wayne International Airport’s Roadmap to Success,” 2026. https://revive-environmental.com/transitioning-arff-vehicles-away-from-pfas-fort-wayne-international-airports-roadmap-to-success/ — *“In late 2024, Fort Wayne International Airport (FWA)… two Oshkosh Stryker ARFF vehicles stocked with legacy AFFF… waste… sent to a facility where it awaits destruction using Revive Environmental’s supercritical water oxidation (SCWO) technology.”*

[^R-c25^]: Revive Environmental — “Revive Environmental’s Impact at the 2025 ARFF Symposium,” 2026. https://revive-environmental.com/revive-environmentals-impact-at-the-2025-arff-symposium/ — *“representatives from over 30 airports”; “Commercial Airports across United States”; “Chemical Manufacturing Facility”* case entries; site snippet: *“transported 950 gallons of PFAS-impacted foam and shell water.”*

[^R-c27^]: GlobeNewswire — “EPA Administrator Lee Zeldin Tours Revive Environmental’s Columbus Headquarters…,” Jul 7, 2026. https://www.globenewswire.com/news-release/2026/07/07/3323732/0/en/ — *“serving municipal, industrial, and federal clients at its permitted facilities in Columbus, Ohio and Grand Rapids, Michigan.”*

[^R-c28^]: EPOC Enviro — “4never PFAS Remediation,” Jan 8, 2025. https://epocenviro.com/4never-pfas-remediation/ — *“4never is available to the landfill and industrial waste management markets in the United States and parts of Canada.”*

[^R-c30^]: ES&T Water (peer-reviewed) — “Spent Media Management Pathways for PFAS Treatment Applications,” 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12260480/ — *“A dedicated system for nonpotable applications will be commissioned in July 2025 to increase capacity to 27,000 kg per month… service anticipated increased volumes of GAC for ground water remediation and industrial wastewater systems in the Midwest. Large scale capacity fully dedicated to potable-use GAC will be available in the first to second quarter of 2026 in the Midwest and New England regions… west coast… by 2027.” Unverified/low-confidence items flagged:* (a) LeadIQ claims a “December 2025 partnership” between Aquagga and Revive Environmental — no corroborating primary source found; likely conflates their co-participation in the DoD DIU/ESTCP demo cohort (Waste Dive lists both). (b) Wurtsmith AFB Annihilator deployment — planned per MLive (Aug 2023); no confirmation it occurred. (c) NJ southern collection site county inferred from CBS Burlington County dateline, not officially named.

Expanded non-dilutive funding channels — from research/grants_ngo.md

[^N-1^]: UNFCCC/GCF — “Accessing the Green Climate Fund” (SAP up to USD 25M; funding windows) — https://unfccc.int/sites/default/files/resource/3.%20%5BGreen%20Climate%20Fund%5D_NBF_Session%205_GCF_20241129.pdf — 2024-11-29

[^N-2^]: Green Climate Fund — Other funding options (REDD+ and MSME pilot RFPs ended) — https://www.greenclimate.fund/access-funding/other-funding-options — accessed Sept 2026

[^N-4^]: GEF — Challenge Program for Adaptation Innovation (direct engagement with private-sector innovators; USD 20M across two initial calls; winners implemented with GEF agencies) — https://www.thegef.org/what-we-do/topics/challenge-program-adaptation-innovation — accessed Sept 2026

[^N-5^]: GEF press release — “$20 million investment in climate adaptation innovators” (>USD 40M across 32 projects; +USD 40M to 2026; LDCF/SCCF) — https://www.thegef.org/newsroom/press-releases/gef-announces-20-million-investment-climate-adaptation-innovators — 2024-11-15

[^N-6^]: FundsforNGOs — 4th Call, GEF-7 Non-Grant Instrument Program (max USD 15M/project; debt/equity/guarantees; Partner Agency submission) — https://www2.fundsforngos.org/latest-funds-for-ngos/4th-call-for-proposals-gef-7-non-grant-instrument-program/ — 2021-01-18

[^N-7^]: GEF — Report of the GEF to the UNFCCC COP (GEF-8: USD 196M to blended finance; biannual calls; SCCF-B tech transfer/private sector) — https://unfccc.int/ttclear/misc_/StaticFiles/gnwoerk_static/tn_poznan/077c785fb367458780568d9770f4098e/d358d92f81574ac4bfd894c6642f3cbd.
— 2025

[^N-10^]: World Bank — 2030 WRG 2025 Annual Report (Gazipur wastewater PPP: USD 40M public + ~USD 45M private; A4R reuse alliance; USD 370M supported 2023–25; USD 1.6B mobilized) — https://www.worldbank.org/en/topic/water/publication/2030-water-resources-group-wrg-2025-annual-report — 2026-05-26

[^N-12^]: UNDP India / 2030 WRG — joint press release (global partners incl. Nestlé, PepsiCo, Coca-Cola, Dow, AB InBev; AfDB, IDB) — https://www.undp.org/india/press-release-joint-publication-2030-wrg-world-bank-undp-unveils-innovative-and-sustainable-partnership-and-financing-models-towards-gender-and — accessed Sept 2026

[^N-13^]: IFC — Water sector page (>USD 5B committed/mobilized since 1994; debt, equity, advisory, concessional finance) — https://www.ifc.org/en/what-we-do/sector-expertise/infrastructure/water — accessed Sept 2026

[^N-15^]: ADB — Technology Innovation Challenge (grant up to USD 450,000 per pilot; 10% co-financing; test/validate in new markets) — https://challenges.adb.org/en/challenges/tic — accessed Sept 2026

[^N-16^]: IDB Invest / PR Newswire — inaugural CHF 100M Blue Bond for sustainable water management in LAC — https://www.prnewswire.com/news-releases/idb-invest-launches-inaugural-swiss-franc-bond-diversifying-investor-base-and-expanding-privatesector-financing-in-latin-america-and-the-caribbean-302786190.html — 2026-06-01

[^N-18^]: IDB — “IDB and IDB Invest provided $26 billion in climate financing over five years” (incl. IDB Lab innovation instruments) — https://www.iadb.org/en/news/idb-and-idb-invest-provided-26-billion-climate-financing-over-five-years — 2022-11-10

[^N-19^]: African Development Bank — GEF partnership page (AfDB as GEF Implementing Agency; >USD 400M secured for member countries) — https://www.afdb.org/en/topics-and-sectors/initiatives-partnerships/global-environment-facility-gef — 2026-06-05

[^N-20^]: OECD — Caribbean Development Dynamics 2025 (debt-for-climate-resilience swap w/ EIB & IDB guarantees tied to sovereign water-security project) — https://www.oecd.org/content/dam/oecd/en/publications/reports/2024/12/caribbean-development-dynamics-2024_86de96a9/a8e79405-en.pdf — 2024-12

[^N-22^]: Bureau Veritas webinar — UNEP-GEF project (EUR 8.8M+) on POPs/PFAS reduction in textiles (Bangladesh, Indonesia, Pakistan, Viet Nam); innovation ecosystem — https://bureau-veritas.easywebinar.live/registration-443 — accessed Sept 2026

[^N-24^]: Josh’s Water Jobs — UNICEF RFP/2025/01/9198242 climate-resilient WASH, Kerala (UNGM registration; e-submission) — https://www.joshswaterjobs.com/jobs/330731/ — 2025-07-02

[^N-26^]: Access Newswire — “WaterEquity Announces $100M in Financing Available for Water and Sanitation Infrastructure” (open application; tickets $2–15M equity/mezz; wastewater treatment/reuse and C&I projects; WaterConnect project prep) — https://www.accessnewswire.com/newsroom/en/banking-and-financial-services/funding-for-growth-waterequity-announces-100m-in-financing-available-942405 — 2024-11-14

[^N-27^]: Xylem/WaterEquity — Water & Climate Resilience Fund >$100M raised of $200M target (Microsoft anchor; Starbucks, Xylem strategic; Ecolab, Reckitt, Gap) — https://www.xylem.com/el-gr/about-xylem/newsroom/press-releases/waterequity-

announces-$100m-raised-for-new-impact-fund/ — accessed Sept 2026

[^N-29^]: Nonprofit Pro — charity: water grantmaking (>$80M/yr in grants; funds local partner organizations only) — https://www.nonprofitpro.com/post/grant-writing-101-charity-water-shares-3-strategies-it-uses-to-effectively-fund-its-programs/ — 2022-01-12

[^N-30^]: Josh's Water Jobs — WaterAid Strategic Partnership Development Manager (secures corporate donations ≥£100K/yr — funding flows to WaterAid, not from it) — https://www.joshswaterjobs.com/jobs/379141/ — 2026-06-25

[^N-31^]: GreenEconomy.Media — Coca-Cola RAIN South Africa ($1.25M grants to five catchment projects, 2019; replenishment focus) — https://greeneconomy.media/coca-colas-rain-set-to-replenish-15-billion-litres-of-water-over-the-next-decade/ — accessed Sept 2026

[^N-33^]: Coca-Cola — Global Water Center East Africa project: US$1.1M in grants from Grundfos Foundation and The Coca-Cola Foundation — https://www.coca-cola.com/ke/en/media-center/east-africa-water-project-launch — 2025-03-20

[^N-34^]: Procurement Magazine — AB InBev 100+ Accelerator (up to USD 100K per startup pilot; water stewardship challenges; 116+ companies) — https://procurementmag.com/articles/what-is-ab-inbevs-100-accelerator-programme — 2024-01-31

[^N-35^]: Startup Grants India — 100+ Accelerator profile (cross-corporate: AB InBev, Coca-Cola, Colgate-Palmolive, Danone, Mondelēz, Unilever; 240+ companies, 40+ countries) — https://www.startupgrantsindia.com/providers/100-accelerator — accessed 2026-07-10 snapshot

[^N-38^]: Food Digital — Diageo water security (Spirit of Progress; up to £5M/5yr Scotland peatland restoration; replenishment) — https://fooddigital.com/news/diageo-promote-water-security-drinks-sector — 2026-02-20

[^N-40^]: CEO Water Mandate Water Action Hub — Suntory Natural Water Sanctuaries ("Mizu To Ikiru"; forest groundwater recharge at plant catchments since 2003) — https://wateractionhub.org/projects/1128/d/suntory-natural-water-sanctuary-okudaisen-and-kasara/ — accessed Sept 2026

[^N-41^]: Ecolab — 2025 U.S. Regional NPO Grant Program guidelines (501(c)(3) required; four focus areas incl. environment/water) — https://www.ecolab.com/-/media/Ecolab/Ecolab-Home/Documents/DocumentLibrary/CommunityRelations/NPO-Grants/Regions/2025/Ecolab-Apply-for-Grant-Information-Regions-for-web-posting-APPLICATION-OPEN-2025-pdf.pdf — 2025-05-29

[^N-43^]: Xylem — Watermark page (nonprofit partners: UNICEF USA, Planet Water, Mercy Corps, EWB; community grants to 148 nonprofits in 2023) — https://www.xylem.com/it-it/watermark/ and https://xyleminc.gcs-web.com/news-releases/news-release-details/xylem-teams-partners-and-communities-globally-accelerate — accessed Sept 2026

[^N-44^]: Xylem — Partnerships Accelerator (TRL ≥7, ≥1 paying customer, invitation-only, applications ~October; 12-month Go/No-Go commercial pathway) — https://www.xylem.com/en-bg/about/xylem-innovation-labs/partnerships-accelerator/ — accessed Sept 2026

[^N-46^]: PMC/World Pumps — Grundfos Foundation overview (DKK 200M donations 2020; Water/Research/Inclusion; NGO partners Water Mission, UNICEF, Oxfam, Plan) — https://pmc.ncbi.nlm.nih.gov/articles/PMC7180044/ — accessed Sept 2026

[^N-48^]: UNDP BIOFIN FIRE database — Quadrature Climate Foundation (eligible: nonprofits/NGOs; nomination process) — https://fire.biofin.org/single/quadrature-climate-foundation — updated 2025-01-09

[^N-50^]: KR Foundation — How we work (no unsolicited applications; invitation-only) — https://krfnd.org/how-we-work/ — accessed Sept 2026

[^N-51^]: Fund for Bangladesh — KR Foundation grant criteria ("does not support for-profit organizations"; DKK 2–5M grants) — https://fundforbangladesh.wordpress.com/2018/09/15/ — 2018-09-15

[^N-52^]: Hinchilla — ClimateWorks Foundation (no open application; periodic targeted RFPs; eligibility has included private enterprises) — https://www.hinchilla.com/funders-us/262303250-climateworks-foundation — accessed Sept 2026

[^N-53^]: AndPurpose — ClimateWorks Adaptation and Resilience Fund RFP (USD 9M pool; deadline Sept 19, 2025) — https://andpurpose.world/climateworks-foundation-adaptation-and-resilience-fund-for-communities-facing-climate-risks/ — 2025-08-25

[^N-54^]: OpenGrants — Deep-tech non-dilutive funding guide (Breakthrough Energy Fellows Cohort 7: Innovator up to $500K via SAFE; Explorer $50–250K grants TRL 0–2; 500Mt CO2e screen; next deadline Nov 2, 2026) — https://opengrants.io/guides/deep-tech-non-dilutive-funding/ — 2026-09-23

[^N-55^]: Inside Philanthropy — Breakthrough Energy Discovery (Fellows, Workshops, Ecosystem Grants; philanthropic funding for early-stage climate tech) — https://www.insidephilanthropy.com/find-a-grant/grants-b/breakthrough-energy — 2026-07-01

[^N-60^]: The Daily Pennsylvanian — Water Center at Penn launches Water-AI Nexus Center of Excellence with AWS, WEF, Leading Utilities of the World (Climate Week NYC) — https://www.thedp.com/article/2025/10/penn-amazon-water-ai-center-climate-change-ai — 2025-10-03

[^N-61^]: WEF/WERF — LIFT program booklet (Technology Evaluation Program; utility-industry vetting system) — https://www.wef.org/globalassets/assets-wef/2-resources/topics/a-n/innovation/technical-resources/ipbooklet15.pdf — accessed Sept 2026

[^N-64^]: NEBRA/WEF — Call for innovative wastewater technologies, LIFT Technology Scan (resource-recovery intensification forum) — https://www.nebiosolids.org/resource-recovery-forum-2015 — accessed Sept 2026

[^N-65^]: Funds for Individuals/NGOs — EREF Research Grants RFA (priorities incl. emerging contaminants in leachate disposal/treatment; typical ≤$300K, ~2 yrs; Dec 1, 2024 deadline cycle) — https://fundsforindividuals.fundsforngos.org/grants/request-for-applications-eref-research-grants/ — 2024-11-18

[^N-66^]: Rotary Club of Port of Spain — Rotary grants overview (global grant minimum budget $30,000; World Fund $15,000–$200,000; DDF matched 100%, cash 50%) — https://rotaryclubpos.org/Stories/what-you-should-know-about-rotary-grants — accessed Sept 2026